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Work Package 6

Early estimates of economic indicators

Deliverable 6.6

Report about the impact of one (or more) big data source (and other) sources on economic indicators

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1 Summary

The main achievement of WP6 was to show that how a combination of (early available) multiple big data sources, administrative and existing official statistical data could be used in creating an existing or new early estimates for official statistics.

Here we provide examples how a combination of (early available) multiple big data sources, administrative and existing official statistical data could be used in creating existing or new early estimates for official statistics. We show that NSIs can shorten the publication lag in a straightforward way without compromising.

The study describes calculated concrete estimates for economic indicators with impact and quality assessment of results from the pilots carried out during the SGA-2. The study describes the implementation of several business cases in particular:

- Big data sources and statistical areas where used
- Other statistical data sources combined with investigated big data sources
- Methods used and impact on quality of results
- Data treatment and related methodological and IT issues

Statistics Slovenia (SURS) and Statistics Finland both explored the usage of traffic loops data in the process of estimating early economic indicators.

Based on investigation of various big data sources SURS had the idea to use the data acquired from traffic sensors and use them as primary and secondary regressor in a linear regression method for nowcasting GDP 45 days after the reference period. The pre-treated traffic loops data have been used as a secondary regressor in a nowcasting method. The method consisted of a linear regression model with principal component analysis (PCA) that was used to find the best fit of quarterly enterprise production data onto GDP values. The study describes the process of data preparation and analysis of the calculated estimates. It was concluded that quite a lot of errors arise from specific characteristics of given datasets (metadata errors, weighting circumstances, editing) and cannot be generalized to the whole or at least to a big part of the Big Data field. The solutions to most of these individual problems need to be specifically made for them. Using of linear regression and PCA methods together with traffic loops data can produce accurate estimations of some early economic indicators.

Statistics Finland carried out extensive testing on methodologies that are capable of dealing with large dimensional data, which are relevant when one tries to extract information from big data and disaggregated microdata that can be used in producing early estimates. Build upon the results obtained from the SGA-1 they explored machine learning approaches to nowcasting Finnish economic indicators (TIO and GDP) using firm-level data and traffic loops. The study describes the process of data preparation and analysis of the calculated estimates. It was concluded that using incomplete early micro-level sources and a real-time big data source such as the traffic loop data can be used to produce early estimates of economic indicators. Statistics Finland regularly produces turnover index nowcasts based on the methodology presented here. Therefore, as a result of this work, it is expected that turnover indicators will become timelier and (what is remarkable) more accurate, which in turn makes it possible to produce the aggregate output indicators (TIO and GDP) faster and more accurately by the existing national accounts procedures. New methodology is provided by which an early estimate of GDP can be produced reliably (quantitatively with similar

accuracy as the official first estimates) by employing micro-level turnovers and traffic loop data together with various machine learning methods designed to handle such large dimensional data.

Through an institutional collaboration agreement with the Bank of Italy, Istat had the possibility to access new data sources concerning electronic transactions of the exchange circuits and interbank settlement of the System of Payments and others deriving from the Anti-Money Laundering Aggregate Reports (SARA) that financial intermediaries file monthly to the Financial Intelligence Unit (UIF), which is a separate branch of the Bank of Italy. A joint working group Bank of Italy-Istat was set up to produce new series of indicators from these Big Data sources, to study the characteristics of these series and to test their possible use besides traditional economic series, for the purposes of nowcasting or forecasting macroeconomic aggregates. This report is aimed at presenting the actions taken to produce the new series and the first results of the insertion of the new series in Istat models. There are two case studies: the nowcasting of the value added of Services and the forecasting of Consumptions. The results are to be considered still preliminary and experimental.

Statistics Poland nowcast the ILO unemployment rate, which is calculated on a quarterly basis and released with a delay of four months. The structural time series (STS) model has been used to produce flash estimates of the ILO unemployment rate, i.e. trend and seasonality. Two variables were considered: (1) registered unemployment rate based on data from District Labour Offices and (2) job vacancy barometer based on online job offers. In the project the following data sources were used: the Labour Force Survey, the registered unemployment rate and online job offers. It was concluded that the job vacancy index actually worsens the ILO unemployment rate. The model without any variables seems to be sufficient but this is mainly because of a clear falling trend in the unemployment rate.

Statistics Portugal aims at assessing early estimates for macroeconomic variables of interest over the most recent reference period based on Big Data in the regular production of official statistics. Regression and Spatial Panel Data Models are used as the methodological approaches.

In this report we show how NSIs can address a major quality issue, namely the timeliness, by using a range of micro level data sources accumulated in the registers well before the official release is made, by employing large dimensional econometric models, to form an initial quick estimate of the target indicator. With the nowcasting we are able to publish preliminary results for GDP well before the official release.

We have shown that this does not necessarily lead to too large revisions, but adds significantly to the quality of official statistics through timeliness dimension.

This line of work can proceed in multiple directions:

- Other data sources can be explored with the methodologies we have presented and possibly in relation to other indicators
- Other modeling frameworks are possible
- A real-time application can be programmed, especially relying on the traffic loops data
- These methodologies can be implemented into the production systems of official statistics, and their added value is not limited to nowcasting the GDP or some aggregate final indicators, but could be explored in order to impute some missing components of the aggregated figures.

Such approaches can be easily implemented across the entire European Statistical System.

2 Estimating early economic indicators using traffic loops data at SURS

2.1 Introduction

The Statistical Office of the Republic of Slovenia calculates the statistics of gross domestic product (GDP) every quarter of a year. 70 days after the reference period is unfortunately the quickest we can publish such statistics as the timeliness of GDP data is limited by the survey evaluations of some of the components that make up GDP. However, the use of flash (rapid) estimates could speed up this process. On the basis of investigation of various big data sources we had the idea to use the data we acquired from traffic sensors and use them as primary and secondary regressor in a linear regression method for nowcasting GDP 45 days after the reference period.

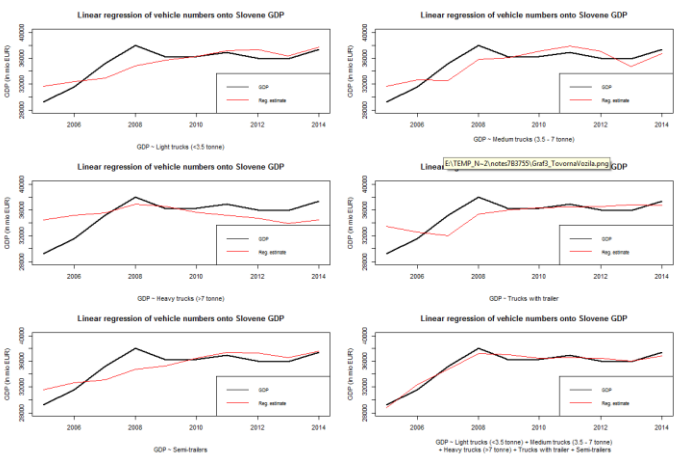
Traffic sensors data was chosen, because we believe that these data have a good correlation, especially if only the subset, representing cargo vehicles on regional roads, is used. Our assumptions are that cargo traffic moves most of the goods across Slovenia and is the main logistic choice behind import, export and goods acquisition for industries and shops. As such it could reasonably be used as a connection between these unobserved quantities and our target variables. We aim to exploit this connection.

2.2 Data sources

Many big data, statistical and other administrative sources could be linked to early economic indicators. As one of the results of the SGA-1 the list of possible sources which could be combined for the purposes of estimates of early economic indicators was prepared. Some of the sources have been already investigated, for some of them there is the issue of their accessibility. Availability of time series of a certain data source should also be taken into account due to our goal to nowcast economic indicators.

One of the data sources which can be easiest acquired is traffic sensor data. Additional advantage is availability of times series of traffic data which is not the case for many other big data sources. First results which have been obtained at the Statistical Office of the Republic of Slovenia (Picture 1) show quite a fit between curves which show movement of annual GDP and movement of estimates of annual GDP based of various annual aggregates of traffic density in Slovenia using data in the 2005-2014 period. The model used for estimation was simple linear regression.

Picture 1: Estimates of annual GDPs using traffic sensors data



From Picture 1 it could be seen 5 examples of estimate of annual GDP due to the aggregated categories of vehicles on Slovenian roads. Categorises which have been used are:

- Light trucks (up to 3.5 t)
- Medium trucks (3.5 -7 t)
- Heavy trucks (more than 7 t)
- Trucks with the trailer
- Semi-trailers

At the example at the bottom right all categories of vehicles were used as a regressor. Surprisingly, the best results were obtained where all vehicles are taken into account. However, based on initial encouraging results more detailed analysis of traffic loops data has been done.

After some research it has been found that the most[1] of the data can be acquired from the Slovenian Ministry of Infrastructure (and municipalities for local traffic). They gave us multiple choices for the format of data and they also provided us with a sample of microdata. Samples of raw data were divided into raw and "so-called" edited data. Raw data present counted categories of counted vehicles per traffic loop, while edited data represent time series of data of one or more traffic loops which were places at the same location point.

2.2.1 Coverage

In the Slovenian case, the coverage of obtained sensors' data is complete. As it is obtained by the Slovenian Ministry of Infrastructure, all existing traffic sensor data are accessible. Furthermore, the list of all traffic sensors on Slovenian roads is available freely online on the Ministry's internet pages, including the geo-position of every sensor. This is a .pdf file, but an excel file can be received with a request to the Ministry. The sensors are present throughout the country on every speedway and regional road; however, it must be said that the number of sensors on the speedways is lacking. Fortunately for us, we have found that Slovenian speedway sensors are not appropriate for economic indicator estimations.

A case for over-coverage could be made from the fact that we are not able to differentiate between foreign and domestic traffic by these data. If this difference is found to be important, some kind of linkage with border traffic sensors will be needed to assess the number of foreign vehicles.

2.2.2 Comparability over time

In the Slovenian case, all existing traffic sensor data are obtained by the Ministry of Infrastructure and therefore can be expected to be consistent and available for a long time. Unless the nature of estimated economic indicators changes in such a way that traffic no longer impacts it, it is reasonably acceptable that this type of data is comparable over time.

2.3 Data handling

Data were acquired from the Ministry of Infrastructure in bulk for now. The first dataset we received was yearly tables of traffic through 647 sensor spots on Slovene regional and speedway roads. The counting intervals are 15 minutes long. The dataset included data from the start of 2011 to the end of 2016. This dataset was acquired in May 2017. The second dataset was received in February 2018 for the data of 2017 in the same format as above. In the future we have plans to move towards a

[1] The data we are talking about here refer to national roads of the regional and highway variety. Additionally, municipality road sensors can be acquired at the administrative offices across Slovenia. SURS also has access to sources which include traffic sensors at national borders.

timelier transfer process, but for the purposes of analysing and using the data in an experimental statistic, we deemed these transfers enough. The 15-minute intervals are the shortest intervals available at the Ministry.

The data we acquired were partly edited already: the actual files contained aggregated numbers of multiple sensors on the same stretch of road if they existed. We have named these objects count spots.

2.3.1 Data editing

The data that each count spot represents are organized into 2 channels: each way on a road is represented by a channel. However, the sensors have the ability to count on two lanes at most; therefore, the sensors on multilane speedways are only able to count one way. Where available, count spots contain both lanes; however, some sensors only count the traffic in one way and the data for the other way are non-existent. Their count spots are filled in zeros for the corresponding channel. All of this has to be taken into account when analysing data, and has to be reliably detected.

2.3.2 Weighting

After analysing the data, we discovered that many count spots lack longer periods of data (such as weeks, months or even whole years). This may be because the count spots were not established yet or they were malfunctioning and turned off for a period. We decided upon using only those count spots that contain at least 85% of the whole date and have no missing years. Our weights were therefore 0 and 1. As we gain newer data these may change to include newer count spots and exclude old ones.

The errors we considered here are whether excluding count spots in this way may lead to losing information about traffic in part of the country. This was decided to not be an issue on regional roads, where traffic is similar in scale and shape, but might represent problems on speedways which were already under-represented. However, to us this is a minor issue, as it was assessed that the traffic on speedways was not representative of domestic economic indicators, because Slovenia is a transit country with a lot of foreign traffic crossing between different parts of Europe. This might skew the weight of such data when used in indicator assessments. Our assumption was proved true when we discovered that speedways do not play a big part in our chosen target variable assessments and actually more often than not return worse results.

Additional weighting happens in the PCA and linear regression process when combining traffic sensor data with other data sources for the purpose of assessing some target variables. These weightings are fully automated by the process and dependent on the parameters used. The traffic sensors are used in two ways: they can either play the part of main regressors (usually with other data sources), in which case the PCA method selects the most important count spots through their principal component weight; or they can be secondary regressors, where they are summed together to form one time series that is used fully in the regression.

In both cases the regression adds additional weights to the regressors that explain how much they play part in the assessment of the target variable.

2.3.3 Data imputation

Imputations had received a big consideration in our work. As said above, we focused on regional roads and the imputations are tailored towards them. The imputations were used when missing data were detected and only after aggregation to a monthly basis was done. This was done in order to

avoid precision errors due to small scale data (one to two cars more in one period may represent growth of 100% in the mornings, while this may only be 1-2% growth in the peak hours) and to limit the data size on disk to a manageable degree. Our imputations of a count spot data are based on the growth of neighbouring count spots.

The possible errors that we detected or anticipated were errors that happen due to algorithmic errors and errors that occur due to the differing scales of traffic in count spots in comparison to their neighbours' traffic. An extensive analysis of scale and shape of data and performance tests of 4 different algorithms were concluded.

The chosen algorithm was tested on a subset of data with non-missing values, and turned out to be quite accurate. On regional roads the error was less than 5% on an imputation of a full month, while the error on speedway sensor data was a bit higher. But, as mentioned above, the speedway data were not as important to us, as we supposed they contain information impertinent to our measurements.

2.3.4 Metadata handling

These represented quite a lot of problems. Even though we received semi-prepared data, the process of preparation clearly had some issues of their own, as quite some errors occur in the metadata. The first problem was indiscriminate use of spaces as both delimiters in files as actual spaces in variable names. The format of other metadata changed throughout time such as:

- Time periods of traffic counting (our choice of the earliest year of obtained data as 2011 was due to the changing time periods of counting from 10 minutes to 15 minutes which occurred inconsistently in count spots data during the previous years)
- Different notations of one-way counting on speedway roads in the files
- Errors in the time notations (from entirely missing time periods to wrong time periods; e.g. one time interval was given at 41 minutes past a full hour when every other interval was strictly at 0, 15, 30 or 45 minutes past an hour)
- In one instance the variable names were inserted in the middle of data
- Some information was given in non-structured format and had to be extracted using regular expressions (for example types of sensor that make up a count spot; whether the count spot represented regional or speedway sensors, etc.)

2.4 Combining of data sources

When we think of combining of data sources in the traditional statistical production we mostly think of combining them at the micro level. If a common identifier exists, the linking of data is quite straightforward; otherwise, the various record linkage methods are applied in order to derive ID in data set where ID is missing. In the area of big data the issue of combining of different data sets is more complicated. Often the (big) data sources are completely different, so we are not able to employ record linkage techniques or one of the data sets contains unstructured data where we need to employ big data techniques such as machine learning in order to link data.

The other possibility for linking of data sources is linking at the macro level. Here we either try to aggregate all data sets at the level which has a common identifier or we include data in nowcasting models.

Nowcasting is a very early estimate produced for an economic variable of interest over the most recent reference period calculated on the basis of incomplete data using a statistical or econometric model different from the one used for regular estimates. Nowcasts may be produced during the very same reference period for which the data are produced.

Conducting one of the pilots during SGA-1 ESSnet project several nowcasting methods were under investigation. For the first stage (dimensionality reduction), the most promising in sense of practical implementation was Principal components analysis method. The central idea of principal component analysis (PCA) is to reduce the dimensionality of a data set consisting of a large number of interrelated variables, while retaining as much as possible of the variation present in the data set. This is achieved by transforming to a new set of variables, the principal components (PCs), which are uncorrelated, and which are ordered so that the first few retain most of the variation present in all of the original variables. For the second stage (statistical learning), many methods were tested (e.g. linear regression). The idea of statistical learning is to learn from a training set of data and infer the function which is validated on the test set of data and can be used to predict future output from future input.

When nowcasting early indicators with the nowcasting method described above, big data and other sources could be combined:

- as regressors in a nowcast equation $\mathrm{Y} = \alpha_1 \mathrm{X}_1 + \alpha_2 \mathrm{X}_2 + \cdots + \alpha_\mathrm{k} \mathrm{X}_\mathrm{k} + \beta_1 \mathrm{Z}_1 + \beta_2 Z_2 + \cdots + \beta_l Z_l$. Variables $\mathrm{X}_1, \mathrm{X}_2, \ldots, \mathrm{X}_\mathrm{k}$ are primary/main regressors (principal components of given set of (big) data, and variables $\mathrm{Z}_1, Z_2, \ldots, Z_l$ are secondary/additional regressors (aggregated sets of (big) data sources).

Table 1: Example of combined data from industry survey and traffic loops data

	P1003162	P1003164	P1003166	P1003168	P1003170	P002	P003	P006	P010
1	216218	44818,67	71895,67	56055,67	260288,3	1824,333	7714,333	15132	4092,667
2	245734,7	80276,67	60211	39325,67	217410,7	1419,667	10247,67	19597,67	5346,667
3	296705	47200,67	64541	75624,33	269679	1392	9770,667	19843,33	6352
4	231986,3	55985	94126	65412	276388,7	1088	8770	23579	5138,667
5	264973	38550,67	66172,67	59620,67	169702,7	1098,333	7693,667	19336,67	4771,333
6	276598,3	29820,67	40195,67	76714	182335	1050,667	9514	19132,67	6456,333
7	247292	19871,67	64405,33	88768,67	198847,3	1064,667	9937	17248,33	6919
8	272853	16915	93066,67	78192	229577	883,6667	7929,667	15822	4068,667
9	300986,3	18163,67	52228	51534,33	185970,3	775	6209	12693,33	3409,333
10	272550,7	13266,33	61242,67	47449,67	246491,3	922	8151	16869	4837
11	288410,3	NA	74888	73008,67	284151,3	1226,333	9448,667	19202,67	5885,667
12	290073	NA	69280,33	63277	202978,3	1091,333	7900,667	16063,67	3566,551
13	310173	NA	83583	57082	162391,7	788,3333	4983,333	13022	2827,333
14	316162,7	NA	75582,67	78835,33	147334	1039,667	6120,333	15912,33	5133,889
15	334938,7	NA	61854,67	57812	262239,3	1191,434	5967,333	16233	5925,667
16	375113,7	NA	84696	56782,67	190403,3	922,3333	5605	15378,67	3462,677
17	342045	NA	72032,33	39791,33	115945	736,6667	5063	13228,67	3653,667
18	334273,7	NA	69265	42082,67	136967,7	957,6667	5777,333	16466,67	5296,667
19	386572,7	NA	65479,67	273164,3	215871,3	1241,667	5768,333	16714	6395,333
20	449406	NA	76241	59811,67	129889	986,3333	5590,333	16164,33	3956
21	404387	NA	37974,33	30490,67	NA	753,6667	4553	13784	3841,333
22	444126,7	NA	78102,67	43745,33	NA	1030,667	5870,667	18748,67	5892,667
23	438757,7	NA	86084	39027,67	NA	1148	5895,667	17713,33	7566
24	492560,3	NA	36321	65785	NA	1196,333	5971,667	17673,33	4992,667

2.5 Estimating early economic indicators

2.5.1 Estimating Gross Domestic Product

After the imputation was done, we have used the final data as a secondary regressor in a nowcasting method. The nowcasting method that consists of PCA and linear regression was used to find the best fit of quarterly enterprise turnover data onto GDP values. Afterward, we predicted the next period with the optimal model.

We have tested many combinations of regressors, by using the nowcasting method that consists of PCA and linear regression with parameters on the main regressors and adding traffic datasets as secondary regressors. We expected that the best result would be obtained when using cargo traffic on regional roads. Since Slovenia is a transit country, the so-called crossroads between East and West and North and South of Europe, our view was that the inclusion of highway traffic would be detrimental to our results. In order to justify our assumptions we used a dataset of full traffic data, a dataset of only cargo traffic data, a dataset of traffic on regional roads and a dataset of only cargo traffic on regional roads.

We have then used these linear regression models to nowcast the estimates of GDP in three points in time (namely second, third and fourth quarter of 2016). The results are quite clear; the traffic data improves the quality of estimates in every occasion in comparison of solely use of industry data, while consistently the best PCA condition seemed to be the *80%*[2], which takes just enough first few principal components to explain 80% of the variability of the microdata.

Table 2: Absolute maximum errors for different traffic count spots datasets

PCA method	Without traffic data (in million EUR)	All vehicle categories, all roads (in million EUR)	Cargo vehicle categories, all roads (in million EUR)	All vehicle categories, regional roads (in million EUR)	Cargo vehicle categories, regional roads (in million EUR)
75%	129.04	151..68	136.85	194.45	101.33
80%	149.46	154.75	146.71	105.55	42.58
85%	200.13	228.54	207.52	207.62	167.30
90%	131.46	232.92	136.62	204.39	106.43
po10	282.41	313.32	248.34	216.84	207.59
zadnja5	102.70	110.33	113.10	118.91	56.78

Table 3: Absolute mean errors for different traffic count spots datasets

[2] The PCA method is used because the number of variables in microdata is too large for linear regression and therefore needs to be reduced. In our models the names *75%, 80%, 85%* and *90%* mean PCA conditions that take just enough first few principal components (variables) to explain 75%, 80%, 85% or 90% of the variability of the microdata. The chosen principal components are eigenvectors with the highest eigenvalues of the covariance matrix of the microdata.
The PCA method with PCA condition *po10* takes just enough first few principal components to have at least 10 times more time points from the beginning to the ending of the time span.
The PCA method with PCA condition *zadnja5* takes all eigenvectors (principal components) whose eigenvalues have at least a 5% share among all of the eigenvalues.

PCA method	Without traffic data (in million EUR)	All vehicle categories, all roads (in million EUR)	Cargo vehicle categories, all roads (in million EUR)	All vehicle categories, regional roads (in million EUR)	Cargo vehicle categories, regional roads (in million EUR)
75%	108.44	120.71	111.17	111.67	83.94
80%	85.47	88.94	88.26	68.65	24.14
85%	115.30	134.35	119.59	150.90	111.14
90%	49.38	82.96	51.11	143.69	68.356
po10	255.25	225.39	211.97	144.45	174.05
zadnja5	66.96	72.78	66.87	48.17	37.20

As can be seen from Table 2 and Table 3, our assumptions about traffic data were right. This can be seen by the fact that in almost every instance any use of traffic data corrects the estimates a little. However, we were pleasantly surprised when we checked for the best combination. Use of any model with the exclusion of *90%* resulted in better accuracy when using cargo data on regional roads. The errors between the official values of GDP and our estimates are on average reduced by a factor of 4, comparing it to the non-traffic example. Furthermore, the maximum absolute error was around 2.5 times smaller when using traffic data in comparison to when not using traffic data. Even the best model without usage of traffic data was still worse than some of the models with included traffic data (but it must be said that we suspect that there was a high chance of overfitting in the best non-traffic model). Lower errors when comparing all roads to regional roads or all vehicle categories to cargo vehicles also support our assumptions.

As can be seen from Table 4 below, the relative errors of estimations when using traffic data are seldom more than one percent.

Table 4: Estimates and errors of no-traffic data and traffic data sets

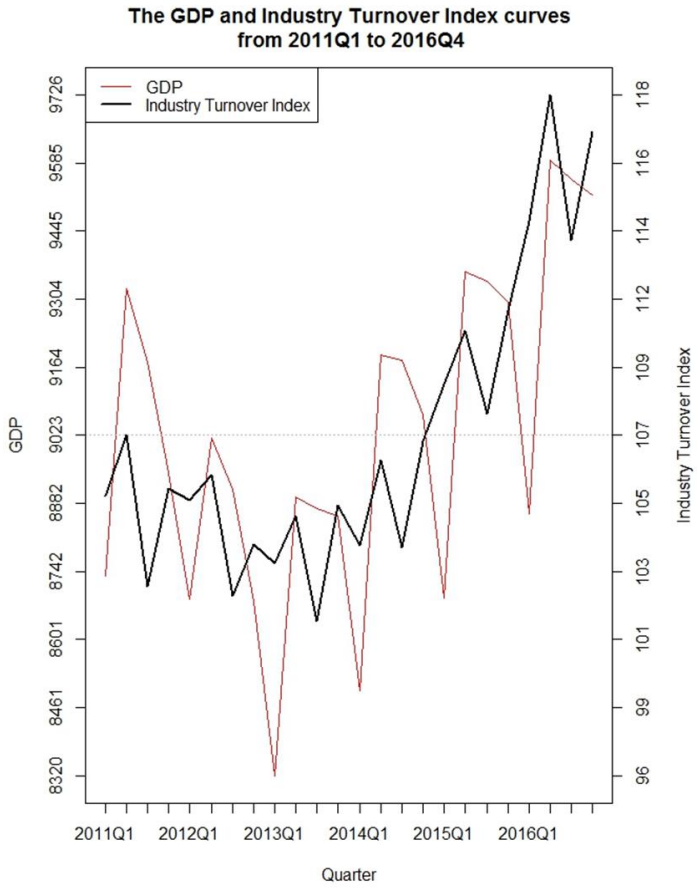
Period	Official values of GDP (in million EUR)	PCA method	No traffic data estimates	Traffic data as secondary regressor estimates	Absolute values of relative errors of the 1st est. (in %)	Absolute values of relative errors of the 2nd est. (in %)
2016Q2	9725.868	*75%*	9596.824	9640.344	1.33	0.88
		80%	9576.406	9708.404	1.54	0.18
		85%	9525.735	9614.310	2.06	1.15
		90%	9594.405	9630.760	1.35	0.98
		po10	9523.429	9548.396	2.08	1.83
		zadnja5	9678.580	9713.673	0.49	0.13
2016Q3	9682.643	*75%*	9613.220	9617.677	0.72	0.67
		80%	9630.709	9640.059	0.54	0.44
		85%	9632.367	9628.078	0.52	0.56
		90%	9693.590	9679.113	0.11	0.04
		po10	9400.231	9545.563	2.92	1.42
		zadnja5	9631.749	9640.023	0.53	0.44
2016Q4	9647.458	*75%*	9520.605	9546.125	1.32	1.05
		80%	9702.478	9635.077	0.57	0.13

		85%	9551.981	9480.159	0.99	1.73
		90%	9653.184	9541.028	0.06	1.10
		po10	9366.567	9439.871	2.91	2.15
		zadnja5	9544.756	9590.680	1.07	0.59

2.5.2 Estimating Industry Production Index

While these are really promising results, we are aware that an assessment on three temporal points is far from optimal. This is why we searched for a longer time series where traffic data could be used. With the assumption that industry production data represent a big part in the evaluation of GDP and therefore the two might correlate well, we tried using the traffic data as primary or secondary regressors for Industry Production Index nowcasts. Encouraged by the results above we expected that traffic data could give good assessments.

Picture 2: GDP and Industry Production Index time series



Like before we tested different linear regression with different PCA conditions. Furthermore we also used different sets of data for regression variables. We decided on testing industry data as fitted with industry data alone, industry and traffic data as primary regressors, industry data as primary and traffic data as secondary regressors and traffic data alone as primary regressors. The learning phase was conducted on the data from 2011 to 2014 (64 temporal points) and the testing phase was conducted on the remaining 24 temporal points. With a larger pool of results we could also improve our criteria for an optimal model. We decided to choose the best model according to the Root Mean Squared Errors (RMSE), which takes into account the accuracy of the estimate in every testing temporal point and also their variability. Therefore the best model is such that has the best

combination of closest average to the true values and the minimal deviation from that value in every temporal point in its monthly estimates in a pre-agreed period.

The actual procedure was to test each parameter at the end of every year, and use the acquired optimal setup when nowcasting monthly values in the following year. An example of this will be given with the next table.

Judging from the RMSFEs of the models (in Table 5), the results are the same as in the above testing. In terms of data combinations, the best appear to use both the industry and traffic data. Meanwhile the best PCA conditions seem to be either 70% or 80%. The zadnja5 model is also close to be the optimal model.

The following table shows the RMSE of tested models:

Table 5: Root Mean Squared Errors for some nowcasting models

	PCA: *70%*; Primary: Industry, Traffic	**PCA: *70%*; Primary: Industry; Secondary: Traffic**	**PCA: *80%*; Primary: Industry, Traffic**	**PCA: *80%*; Primary: Industry; Secondary: Traffic**	**PCA: *90%*; Primary: Industry, Traffic**	**PCA: *90%*; Primary: Industry; Secondary: Traffic**	**PCA: *zadnja5*; Primary: Industry, Traffic**	**PCA: *zadnja5*; Primary: Industry; Secondary: Traffic**
2015	2.90	2.50	2.25	2.84	4.45	5.97	2.82	2.47
2016	2.25	2.74	2.03	3.28	3.58	3.11	4.14	2.94

Based on the results of testing, in both years we would pick the model with the PCA condition 80% and both sets of data as primary regressors as our optimal model. The actual assessment of the Industry Production Index would be produced in the following way:

1. At the end of 2015 we would test different models and according to the results choose the optimal model (PCA: 80%, both sets of data as primary regressors).

2. We would use this model for every monthly nowcast of 2016 and declare the assessments as our flash estimates.

3. At the end of 2016 we would again test all the models and choose the new optimal model (in this case the same one) to be used throughout 2017.

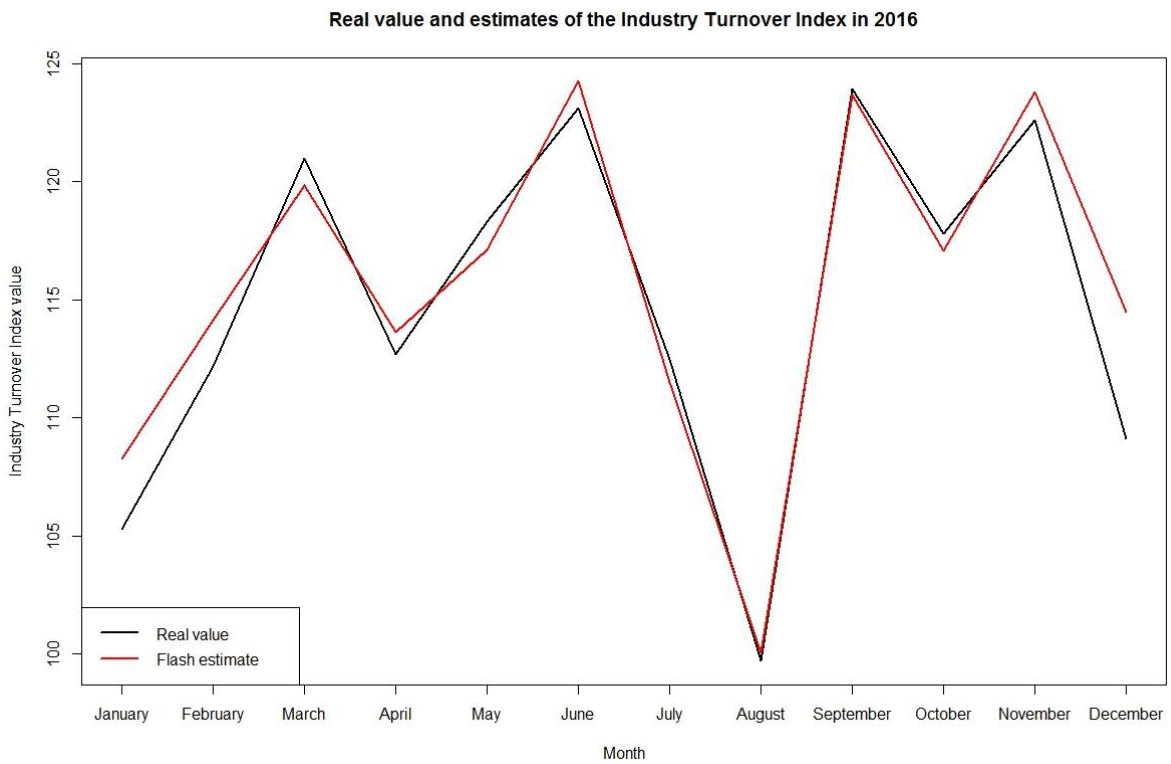
Using this strategy our estimates for the indices compared to their real values in the months of 2016 are:

Table 6: Real values and estimates of the Industry Production Index in 2016

Month	Real value of the Industry Production Index	Flash estimate of the Industry Production Index	Absolute Errors	Relative Errors (in %)
January	105.3	108.27	2.97	-2.82
February	112.2	114.12	1.92	-1.71
March	121.0	119.82	1.18	0.98
April	112.7	113.63	0.93	-0.83
May	118.3	117.11	1.19	1.01
June	123.1	124.25	1.15	-0.93
July	112.5	111.51	0.99	0.88
August	99.7	100.05	0.35	-0.35
September	123.9	123.68	0.22	0.18
October	117.8	117.08	0.72	0.61
November	122.6	123.79	1.19	-0.97
December	109.1	114.47	5.37	-4.92

We can see that in most cases the relative errors are in the 1% limit, and the largest relative error is less than 5% off of the true value. This can be seen even better on the graph of comparisons of the two series below.

Picture 4: Comparison of the real values and estimates of the Industry Production Index in 2016



2.5.3 Continuation of estimating with newer data

Obtaining new data from 2017, we have tried to reproduce these results. The overall experiment was conducted on additional 4 time periods – every quarter of 2017. Once again the estimations were calculated using a PCA method with five different parameters and using the traffic data in different

ways: either a total sum of traffic data in an interval as a secondary regressor or individual count spots as primary regressors to be trimmed by the PCA method.

The newer results for 2016 and 2017 are presented in the table below.

Table 7: Absolute maximum errors for different traffic count spots datasets

PCA method	Without traffic data (in million EUR)	All vehicle categories, all roads (in million EUR)	Cargo vehicle categories, all roads (in million EUR)	All vehicle categories, regional roads (in million EUR)	Cargo vehicle categories, regional roads (in million EUR)
75%	155.07	266.88	158.51	154.98	135.00
80%	200.08	155.68	135.22	576.83	230.38
85%	324.01	111.88	321.65	419.63	267.41
90%	261.71	221.11	295.35	274.29	318.61
zadnja5	224.39	273.89	269.87	416.65	218.09

Table 8: Absolute mean errors for different traffic count spots datasets

PCA method	Without traffic data (in million EUR)	All vehicle categories, all roads (in million EUR)	Cargo vehicle categories, all roads (in million EUR)	All vehicle categories, regional roads (in million EUR)	Cargo vehicle categories, regional roads (in million EUR)
75%	81.07592	106.34516	102.92959	55.07030	60.96715
80%	50.84946	68.59371	53.96622	26.16266	30.22127
85%	1.605270	53.435054	19.905594	79.987767	113.048431
90%	62.86737	128.62754	120.81747	170.62065	79.05392
zadnja5	62.93502	111.45412	174.36118	47.16852	57.26990

As we can see from these tables, some findings from the previous experiments seem to still hold true. Estimating with all vehicles on all roads gives worse results than using no traffic at all. Likewise, using only cargo traffic on all roads is not the best choice either. From both tables we can see that the *90%* parameter gives bad results, as the maximum error throughout the time intervals is big, and the mean error shows that estimations with non-traffic regressors perform better. However, the estimation is not the best possible. We can also see that in these new periods an alternative to the cargo traffic on regional roads has arisen for good estimation. Using all vehicles on regional roads has the same low mean error, but its maximum errors are quite big compared to the cargo traffic on regional roads.

In the end it seems like the best series is still the cargo traffic on regional roads series. Just as with the previous data, we directly compare official values to the values of estimation without and with traffic data. The official values 2016 are again included, since the GDP series is subject to revisions and the values are different.

Table 9: Estimates and errors of no-traffic data and traffic data sets

Period	Official values of GDP (in million EUR)	PCA method	No traffic data estimates	Traffic data as secondary regressor estimates	Absolute values of relative errors of the 1st est. (in %)	Absolute values of relative errors of the 2nd est. (in %)
2016Q2	9755.848669	*75%*	9641.769	9620.851	1.17	1.38
		80%	9655.510	9601.865	1.03	1.58
		85%	9627.988	9660.667	1.31	0.98
		90%	9689.009	9531.461	0.69	2.30
		zadnja5	9531.461	9541.131	2.30	2.20
2016Q3	9748.845959	*75%*	9642.126	9699.763	1.10	0.50
		80%	9681.626	9637.750	0.69	1.14
		85%	9656.819	9658.668	0.94	0.93
		90%	9716.181	9464.554	0.34	2.92
		zadnja5	9704.672	9530.760	0.45	2.24
2016Q4	9686.031852	*75%*	9570.723	9700.913	1.19	0.15
		80%	9629.284	9582.602	0.59	1.07
		85%	9722.534	9553.433	0.38	1.37
		90%	9593.896	9600.487	0.95	0.88
		zadnja5	9646.320	9646.767	0.41	0.41
2017Q1	9395.205718	*75%*	9355.285	9317.238	0.43	0.83
		80%	9419.886	9336.230	0.26	0.63
		85%	9285.980	9305.568	1.16	0.95
		90%	9133.499	9275.574	2.79	1.27
		zadnja5	9300.724	9308.459	1.01	0.92
2017Q2	10197.88759	*75%*	10137.822	10103.581	0.59	0.92
		80%	10201.657	10118.330	0.04	0.78
		85%	10111.837	10096.056	0.84	1.00
		90%	10130.813	10178.251	0.66	0.19
		zadnja5	10182.237	10248.388	0.15	0.50
2017Q3	10187.24812	*75%*	10151.038	10077.054	0.36	1.08
		80%	10164.924	10045.334	0.22	1.39
		85%	10148.311	10002.641	0.38	1.81
		90%	10152.910	10505.859	0.34	3.13
		zadnja5	10273.918	10347.609	0.85	1.57
2017Q4	10265.53999	*75%*	10110.472	10224.349	1.51	0.40
		80%	10065.457	10099.572	1.95	1.62
		85%	10346.287	9998.1344	0.79	2.61
		90%	10188.307	10232.035	0.75	0.33
		zadnja5	10277.841	10339.279	0.12	0.72

As the GDP statistic is one that is revised with new data, the official values changed through the year. We are now trying to devise a strategy of estimation around the changing values: common sense dictates that the target variable in the training set should change according to the current information! Since revisions of values happen in every quarter for the current year and in the last

quarter for the previous year, it is important to consider which information is fed to the algorithm at which point in time.

With the new data we were able to create new estimates for the Industry Production Index. As with the GDP series, this too is subject to revisions and the old 2016 values have changed, although not by much. The table below presents the new values for 2016, using the same PCA – linear regression parameters as before: the *80%* PCA parameter and using industry data and count spots as primary regressors to be fed to the PCA method. Following our strategy of calculating the best PCA parameter at the end of a year to use throughout the next year, new estimates for 2017 are also included using the parameters we determined at the end of 2016: again the *80%* PCA method with both industry and traffic data as primary regressors.

Table 10: Estimates for the Industry Production Index

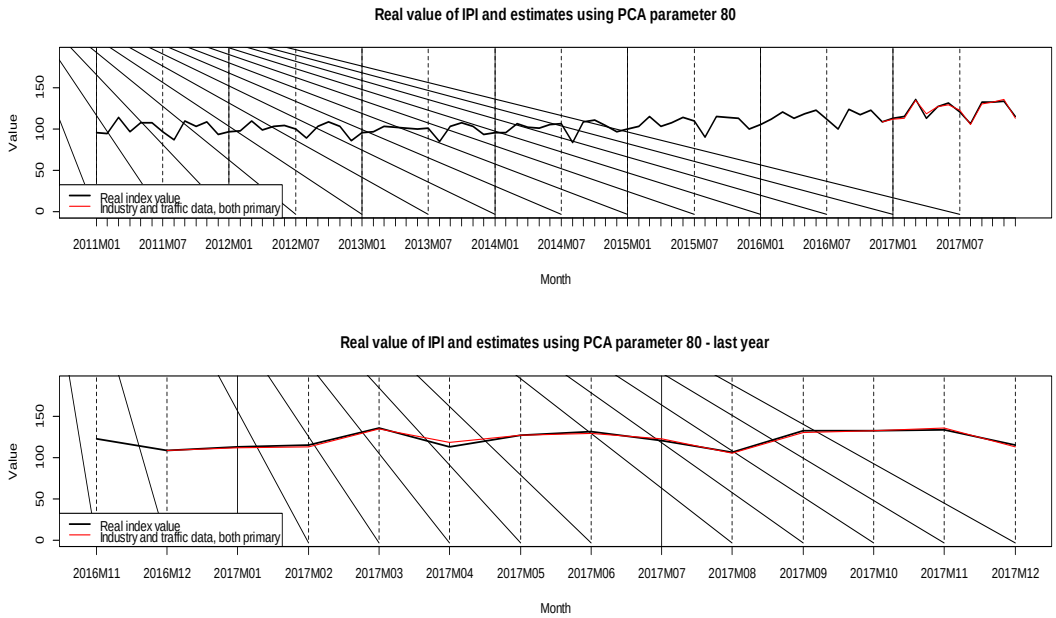
Month	Real value of the Industry Production Index	Flash estimate of the Industry Production Index	Absolute Errors	Relative Errors (in %)
January 16	105.3	102.15	-3.15	-2,99
February 16	112.1	108.34	-3.76	-3,36
March 16	120.9	120.28	-0.62	-0,51
April 16	112.9	112.80	0.20	0,18
May 16	118.1	117.63	-0.47	-0,40
June 16	123.0	119.70	-3.30	-2,69
July 16	112.4	114.07	1.67	1,48
August 16	99.6	98.61	-1.00	-1,00
September 16	123.7	123.98	0.28	0,22
October 16	117.7	121.89	4.19	3,56
November 16	122.4	122.40	-0.00	0,00
December 16	109.0	110.63	1.63	1,49
January 17	112.6	111.95	-0.65	-0,57
February 17	115.6	113.12	-2.48	-2,15
March 17	135.7	135.12	-0.58	-0,43
April 17	113.4	118.32	4.92	4,34
May 17	127.3	127.35	0.05	0,04
June 17	131.8	129.14	-2.66	-2,02
July 17	120.2	122.79	2.59	2,15
August 17	106.8	105.72	-1.08	-1,02
September 17	133.2	130.71	-2.50	-1,87
October 17	132.8	132.16	-0.64	-0,48
November 17	134.0	135.92	1.92	1,43
December 17	115.3	112.68	-2.62	-2,27

As we can see, the estimates are consistently good, with none of them being over 5% off of the real index values. According to our algorithm however, in the next period we are to use a different series for the estimation process. The best estimation according to the RMSFE is the one using cargo traffic on all roads with the *80%* PCA parameter. To check whether this is true, we will need to wait for the end of 2018 to get new data. For now we are pleased with the results, as they continue to inspire confidence in the algorithm and our reasoning.

The reasoning behind these decisions is shown in the following table:

The figure shows how our estimations for 2017 (in red) correspond to the real values of the index (in black).

Picture 5: Real and estimated IPI



2.6 Impact assessment of use of traffic loops data on estimating economic indicators

What we gathered during our processing of data and the errors is that quite a lot of errors arise from specific characteristics of given datasets (metadata errors, weighting circumstances, editing) and cannot be generalized to the whole or at least to a big part of the Big Data field. The solutions to most of these individual problems need to be specifically made for them. Lucky for us the source of traffic sensor is fairly consistent and the solutions we devised in this pilot can be applied in the future as well without (much) changes. Furthermore, some of these problems are sensor-specific and while each problem needs to be addressed individually, they are part of a common set of problems that arise when working with other smart sensors as well. As such a lot of solutions on other smart sensor data can probably be adapted from today's (a good example is missing data due to malfunctions).

We have shown that the use of linear regression and PCA methods together with traffic loops data can produce accurate estimations of some early economic indicators. Additionally, since data acquired by smart sensors such as traffic density are measured automatically, we could replace some of the sources used in the normal calculation of the statistic with these data and reduce the burden of survey respondents, or use it as an additional burden-free source for more accurate calculations.

What still needs to be done: in some years enough quarters will pass to efficiently use RMSFE criteria for optimal model selecting on GDP. At that moment we will be able to accurately assess whether is it better to flash estimate directly GDP or if we should focus on rapid estimations of its less-timely components and then use them as regular data in the normal estimation process for GDP. In addition it has to be investigated what precision is required in order to produce reliable earlier estimates,

explore the possibilities to nowcast only some components of quarterly GDP (for which all data are not available on time), include other big data (and other) sources and also test some additional methods for nowcasting.

Going outside of national statistic territory, traffic data could be researched for use in travel/tourism statistics, population mobility statistics and regular traffic statistics, etc.

3 Early estimates of economic indicators at Statistics Finland

3.1 Background

Statistics Finland carried out extensive testing on methodologies that are capable of dealing with large dimensional data, which are relevant when one tries to extract information from big data and disaggregated microdata that can be used in producing early estimates. As a backdrop to the Finnish pilot, the need for methodological improvements in short-term statistics was pressing, given the user discontent of the late publication and large revisions of short-term indicators. Further, Eurostat is expected to ask for timelier delivery of both monthly turnover indices, and the flash estimate of the quarterly GDP. Statistics Finland has historically had large revisions in the monthly indicators as well as relatively slow dissemination schedule (at t+75 days). The first estimates are produced at t+45 for national accounts and Eurostat but the revisions are very high since the data are not completely accumulated. The reasons for large revisions can speculatively be related to size of the economy (very small) and the presence of very few very large companies which can by themselves create fluctuations in the aggregate figures (as much as 1/3 of total output fluctuations can be explained by idiosyncratic shocks to large companies (Fornaro & Luomaranta (2017)), and accuracy therefore depends on obtaining the survey results especially for the most important companies in time. This is why a sampling scheme that tries to achieve representativeness of the economy will have only limited use. The investigation of the possibility of expanding the survey to consider also the very small firms, reveals that in order to achieve representativeness, one would need to increase the sample size by a very large margin (e.g. the sample should be three times larger), and it is also understood that the responses provided by the smaller firms might not be very reliable. This is why increasing the sample size is not desirable and it is against the current policy of alleviating response burden whenever possible. Because of these considerations, Statistics Finland collects turnovers from the largest companies (around 2000 firms in the sample) in the turnover inquiry (continuously accumulating but fairly complete at t+30), and then waits for the VAT data (t+52) for smaller firms. The schedules in these two data sources create problems for the preliminary estimate at t+45, when the VAT is missing, but even at t+75 when the VAT is not complete as a large share of small firms do not need to report monthly VAT figures to the tax authority. This is the second major reason for large revisions in the Finnish turnover indicators, because the growth rates of small and large firms are known to be very different, especially in economic cycle turning points.

3.2 SGA1: improving the relevance of turnover indicators

For the purposes of reducing revisions and producing early estimates, it makes sense to try to estimate the component which is missing in the data (the firms which would normally be included only when the VAT data are obtained). In the SGA-1, the Finnish pilot conducted tests on turnover indicators with the strategy of estimating the missing part (the firms that report only the VAT or otherwise have not responded to the inquiry). We also test the performance of large dimensional models at very early time periods using the accumulated firm-level data without making the separation of sources.

A starting point was the basic idea of using early responses from the accumulating survey results. We have employed factor analysis, as well as shrinkage regressions to extract information from the accumulating turnover inquiry. The results are well documented in https://www.etla.fi/julkaisut/nowcasting-finnish-turnover-indexes-using-firm-level-data/

The results from producing early estimates based on incomplete data were very well received at the Finnish statistical office, and a regular production of these estimates for testing purposes has been ongoing for almost a year (the data are also transmitted to Eurostat and to national accounts, as of February 2018). What is moreover interesting, that the statistical techniques seem to be able to correct for a major problem in the series, namely the biased first publication by the statistical office. This particular problem can be sourced to the data accumulation from different sources (VAT and survey) in a deterministic fashion. Thus, we have contributed to the timeliness as well as accuracy dimensions in the SGA-1.

We reproduce tables which gives a clear idea of the potential improvement in both accuracy and timelines for the Finnish main industry turnovers. The table below documents the accuracy vs. timeliness trade-off for estimates obtained with different models (Factor models, Ridge, Lasso, and Elastic-Net regressions). The figures reported are MAE in different time periods when the nowcasting model is run, and obviously the later periods have more accumulated firm-level turnovers available from the survey.

In all the cases firm-level data are used as input, while the indicators of the main aggregate industries are the output variable for which we want to learn the next year-on-year growth rate before it is officially available. The results indicate that 1) the shrinkage regressions perform better than factor models, 2) we obtain very competitive estimates for services, trade, and construction industries (for manufacturing the results are less convincing), and 3) all the results we obtained beat consistently the automated ARIMA forecasts (obtained using SAS X12 procedure).

Table 11: Nowcasts at t+5, t+10, t+20, t+26 for main industries turnover, MAE

	$t\|5$	$t\|10$	$t\|20$	$t\|26$
		Manufacturing		
Elastic-Net	2.097	2.324	2.017	1.999
Lasso	2.268	2.369	2.177	2.122
Ridge	**1.833**	**1.861**	**1.712**	**1.675**
Factors	2.739	2.582	2.166	2.036
t|45 (current method)	**1.084**			
		Services		
Elastic-Net	1.471	1.248	**0.979**	**0.984**
Lasso	1.508	1.251	0.980	0.999
Ridge	**1.269**	**1.212**	1.071	1.076
Factors	1.707	1.348	1.183	1.166
t|45 (current method)	**1.730**			
		Trade		
Elastic-Net	1.978	1.697	**1.165**	**1.162**
Lasso	2.045	1.758	1.172	1.175
Ridge	**1.762**	**1.650**	1.292	1.297
Factors	2.272	1.726	1.441	1.443
t|45 (current method)	**0.895**			
		Construction		
Elastic-Net	3.302	3.620	2.851	3.054
Lasso	3.358	3.641	2.800	3.098
Ridge	**2.982**	3.308	3.029	2.988
Factors	3.466	**3.215**	**2.487**	**2.359**
t|45 (current method)	**2.555**			

We then investigated if we can make more accurate estimates of the final series by splitting the population into two parts. The first consists of the firms which are surveyed in the turnover inquiry (the large firms), and the second consists of firms whose turnovers are not yet available and the statistical office would need to wait until the VAT is available at t+52 days (these are the smaller firms). We extract factors from the turnover data which is available at t+28, and use the first few

factors (which correspond to largest eigenvalues in PCA) as extra input variables in an ARIMA model, to forecast the development of the smaller firms, after which we aggregated the forecasted part and the already available part from the turnover inquiry to form final values of the indices. These tests performed remarkably well, beating the current accuracy by a large margin at an earlier point in time, by 15 days. The table below provides the results from testing the accuracy each month over the period of 2 years (2016-2017), which is limited because of some problems in obtaining the realistic data vintages for very early observations. We solve this issue for the SGA-2 when testing similar methods for GDP series.

Table 12: Nowcasts at t+28, where part of the source data is forecasted

	Manufacturing	Services	Trade	Construction
MAE of estimate 1 factor	0.498	0.465	0.580	1.547
MAE of estimate 2 factors	0.485	0.453	0.554	3.121
MAE of estimate no factors	1.006	0.988	0.981	2.201
MAE of $t\|45$ (current method)	1.675	2.161	0.763	2.905

The time series model we construct is based on the US. Census Bureau program X12 which is available in SAS. It can handle seasonality, outliers and missing data, and constructs an optimal time series ARIMA structure for us (even though the fact that we use year-on-year changes alleviates the seasonality issue). We use this program for convenience, in order to test hundreds of time series and nowcasting periods with a relative ease, and are confident of its reliability because it is used in the official statistics for seasonal adjustment procedures. We use the R-package "glmnet" for obtaining results for the shrinkage regressions, and R to do factor analysis (in practice, we choose to extract factors by PCA, since more complicated methods do not appear to provide any improvement in forecasting accuracy in our setting).

3.3 SGA2: machine learning approaches to nowcasting Finnish GDP using firm-level data and traffic loops.

We build upon the results obtained from the SGA-1 to explore many more data sources and methodologies. Having seen that the models which were able to incorporate some firm-level idiosyncratic information (as opposed to estimating the common components as in factor analysis) were performing better than the factor models, we are interested in expanding the set of models that can be applied in our setting. Many machine learning algorithms exist that are designed precisely to deal with high dimensional econometric problems to forming predictions out-of-sample. The algorithms that we had used so far are linear regressions, but more complicated algorithms exist that can add information in a non-linear manner. For example a certain input variable (say a firm) could lose explanatory power over the testing period, or some input variable should only be considered in the upswing of the economy. Regardless, it is interesting to explore if we can find better performing algorithms that can best explain the economic fluctuations and use them in a nowcasting problem. Another important methodological improvement we make is the use of model combinations in line with the known results in machine learning where the predictive ability of the model can be improved by creating model ensembles, i.e. a model whose output is a combination of models with different structures.

Moreover, we expand the data sources by exploring many alternative regressors. Among others, the confidence indices (business and consumer), VAT, supermarket scanner data, wages and salaries, and

traffic loops. We find that the traffic loop data provide a competitive alternative to firm-level turnovers, capturing the essential fluctuations in the economy (see the above section with the Slovenian examples). We extract this real time data source from the Finnish transport agency website, where it is available even at the daily frequency.

I will next give more detailed summary of the data, methodology and results.

3.3.1 Data

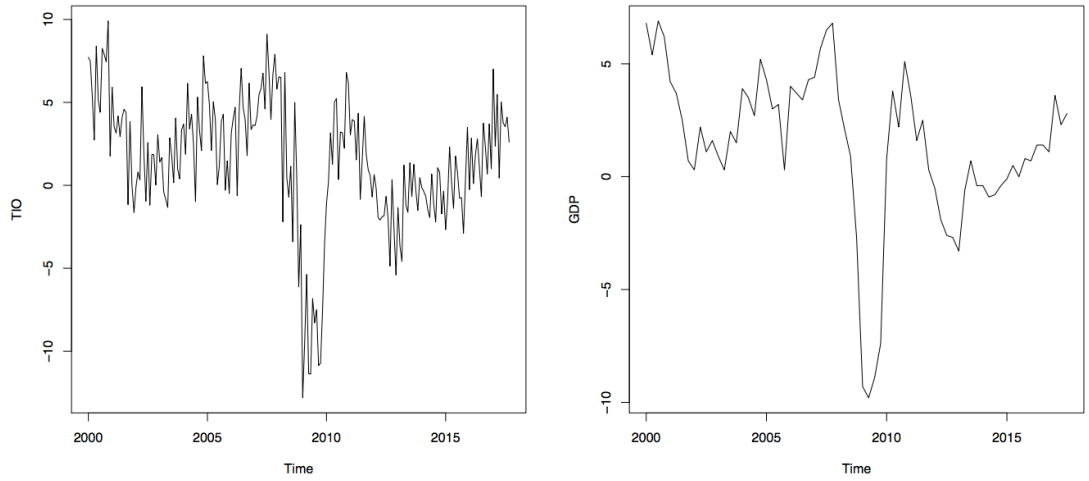
The main regressors in our nowcasting application are firm-level sales extracted from the sales inquiry, a monthly survey conducted by Statistics Finland for the purposes of obtaining turnovers from the most important firms in the economy. This dataset covers around 2,000 enterprises and encompasses different industries (services, trade, construction, manufacturing), representing about 70% of total turnovers. The data are available soon after the end of the month of interest and a considerable share of the final data is accumulated around 15 to 20 days after the end of the reference month. Formally, Statistics Finland imposes a deadline to the firms, which are supposed to send their data by the end of the 15th day of the month. We therefore choose to compute the nowcast on the 16th day. However, this deadline is not always met, thus our set of firms' sales does not cover the entire sample. The data accumulation is realistically simulated by using the time stamp of the reported sales, which allows us to track what data were available by each date of a month. Further, the more recent data points, starting from January 2017, are based on real time data collection.

We also require that firms have long time series (starting in 2006), and that they have reported sales figures by the date we extract their information from the database. By excluding entries and exits, as well as the firms with missing data, we avoid having to carry out imputations, and restrict to a dataset with more stable behaviour. There are a few important enterprises which we know to have the potential of having significant explanatory power, which we individually make sure that they are included in the data. We extract data of the firms that have reported sales by 16 days after the end of the reference month, leading us to have 800 firms on average in the regressors' set. We compute the sales growth rates for all the months from 2006 until the nowcasted month of interest. If the firm has reported sales by the t + 16 at the nowcasted month, but has missing values during the time span (i.e. the firm did not reply at some earlier date, or the firm was not included in the turnover inquiry at that time), we try to obtain the missing growth rates from VAT data, which should include all the firms in the economy.

The target variables in our exercise are the Trend Indicator of Output (TIO) and quarterly GDP, both measured in real-terms year-on-year growth rates. The TIO is a monthly series that describes the development of the volume of produced output in the economy. It is constructed by using early estimates of turnover indices (not publicly available), which are appropriately weighted to form the monthly aggregate index. The TIO is published monthly at t + 45, and its value for the third month of a quarter is used to compute the flash estimate of GDP, which is also published as an early version at t + 45, and updated at t + 60. The t + 60 version is considered as the first official and reliable estimate of GDP. Thus, given the information we have provided, the TIO in fact represents a GDP nowcast in its own right. We stress the importance of using the realistic vintages, as the data are constantly "improved" by many internal processes, and by the accumulation of new data. The usage of revised data can arguably lead to too optimistic views on the nowcasting performance of our models and consequently invalidate the results, especially in the eyes of the statistical office staff members who

we wish to convince about the usefulness of the proposed methodology. The methods need to be working with the realistic data that the official statisticians have to deal with. Below we report the plots of the TIO and GDP year-on-year growth rates.

Picture 6: Monthly and quarterly measures of GDP (TIO and flash)



(a) TIO year-on-year growth, monthly series (b) GDP year-on-year growth, quarterly series

One aspect that it is important to underline is how closely related the TIO and GDP growth are. If we aggregate TIO growth to the quarterly level we obtain a series that closely tracks GDP growth (the resulting correlation coefficient is 0.99). This demonstrates that providing a good estimate of TIO leads to a greater nowcasting accuracy of GDP.

3.3.1.1 Traffic data

One of the main objectives of the ESSnet Big Data Early Estimates working group is to investigate the nowcasting potential of big data sources. The firm-level data which constitute the main data source of our exercise provide a good nowcasting performance, as we are going to show later. However, while high-dimensional, the firm-level turnovers we use are not a big data source in a traditional sense. In particular, they are not collected in real-time, i.e. firms send their data after the end of the reference month. Moreover, our turnover dataset is structured and fairly easy to handle, which is not typical of big data.

In addition to the nowcasting exercise involving firm-level data, we consider the predictive performance of traffic volumes records obtained by the Finnish Transport Agency website[3]. This dataset contains the number of vehicles passing through a number of measurement points (about 500) around Finland, observed through an automatic traffic monitoring system. The data are available at hourly frequency, and it distinguishes between different types of vehicles. This dataset contains numerous missing values, due to the fact that some measurement points do not have observation for certain days or months, and it is not structured. For our nowcasting analysis, we collect data for trucks' traffic volumes from January 2010 (the first dataset available), in particular their year-on-year growth rate. Trucks' traffic presents an interesting intuitive link with aggregate economic activity, especially with trade. We expect that in periods of economic growth, where trade

[3] The data are available at: https://aineistot.liikennevirasto.fi/lam/reports/LAM/ .

volumes increase across the country, we should observe a higher number of trucks' passages, in order to move goods. Of course, this does not cover the transfer of services and other types of economic activities, but it should still present some positive correlation with economic activity growth.

3.3.1.2 Other input data sources

In our study, we have also investigated other data sources, which are briefly mentioned here. We have decided to concentrate the report on the use of firm-level data as regressors due to their timeliness and due to the fact that none of the additional data sources improved our results noticeably (adding additional regressors has usually led to a deterioration of the nowcasting performance, in our setting). Thus our contribution (and this report) will be more contained. As the confidence indices of various types have been shown to predict economic activity, see e.g. Christiansen, Eriksen, and Møller (2014), we also examined the consumer confidence index published by the statistical office, as well as industrial confidence indices published by the Confederation of Finnish Industries, but we find that once they are incorporated amongst regressors, the results do not improve. Data on wages and salaries, which are also timely (available at t + 20), did not improve the nowcasting accuracy either. In addition, we try to improve the coverage of our input data set to smaller enterprises by obtaining the information of small firms' turnovers from VAT data. These VAT figures are available with a considerable delay (usually one month after the sales inquiry), thus we needed to forecast their development to incorporate them in our models. Again, the use of these data, both on their own and in combination with the rest of the regressors, did not improve the nowcasts. Finally, we have quickly explored the scanner data obtained from a large supermarket chain, but unfortunately the data set we worked with did not cover a sufficient number of periods in order to meaningfully add it in the testing procedures.

3.3.2 Formulating the nowcasting exercise

To make sure that the overall nowcasting procedure is feasible in a real-time setting, we need to consider two important aspects: data availability and computational feasibility. The first issue boils down to the fact that, while testing the nowcasting models, the researcher should not rely on data which would not be available in real time. This implies that we have to take into account the publication lag in a realistic fashion. For example, in our application we compute the nowcast at t + 16, i.e. 16 days after the end of the reference month or quarter, thus we should not use data sources which are not available by then (for example VAT data). The other important aspect revolving around data availability concerns the use of the correct vintage of data, which is the one that reflects the information available to the statistical institute at the time the nowcast would have been computed. Most economic series are revised multiple times, both because of estimation error and of benchmarking. The practitioner should avoid using the final value of the indicators of interest, including the target variables (in our case GDP and the TIO) and the regressors (e.g. the firm-level sales in our application), and focus on collecting realistic, non-revised, versions of the indicators.

In our nowcasting exercise, we are careful in terms of making a realistic representation of the available information set. With respect to the target variables, things are rather straightforward. Computing estimates at t + 16 means that we have the previous value of TIO. For example, suppose that we want to nowcast TIO for March: we would compute the nowcast on 16 April and, given the release schedule, at that date we have TIO data for February. We would then estimate a model using data up to February and then compute the nowcast using the March firm-level sales. When

estimating quarterly GDP, we do not rely directly on the GDP series but rather use TIO, which means that we do not have problems in terms of publication lag. Fortunately, we are able to realistically simulate the accumulation of firm-level data, because Statistics Finland records that date on which the firms send their sales reports.

While the publication lags are easy to take into account in our setting, the use of realistic vintages has proven to be somewhat harder to tackle. The TIO is revised multiple times, even many months after its release. Moreover, these revisions do not incorporate solely corrections of the estimate due to the expansion of the data sources but are also affected by benchmarking. This fact implies that if we use the final version of TIO in the estimation and in the evaluation of the nowcasts we would put ourselves in a dramatically different scenario than the one faced in real time by the statistical office. Moreover, our nowcasts would contain errors that are not due to the lack of regressors but that are instead caused by the lack of smoothing and benchmarking. Consequently, we use vintages reflecting the first estimate of TIO and adopt these initial figures as target to evaluate our nowcasts. Unfortunately, the historical vintages for TIO are available only since March 2012, meaning that our nowcasting exercise does not cover some interesting periods such as the Great Recession of 2008–2009. However, we are left with more than 60 predictions to be made and the timespan going from 2012 until 2017 does include periods of high growth and months of considerable output drop. On the regressors' side of things we a have a similar problem, i.e. the firm-sales are revised over time. These corrections include actual revisions made by the firms (even though these adjustments are relatively small) and the corrections for organic growth made by Statistics Finland. In particular, the statistical institute adopts a growth-correction methodology which cleans sales growth caused by mergers and acquisitions. While the timing of these corrections is not clear, we want to avoid being overly optimistic in terms of the data availability at the time of the nowcast, thus we rely on the original, not corrected, version of the firm-level data.

Now to the structure of our empirical exercise: we start to compute monthly nowcasts of TIO from March 2012. In particular, we extract a panel of firm-level sales which starts from January 2006 and contains information until March 2012. Notice that our panel is balanced (i.e. we select firms which are present throughout the time interval of interest). In real-time setting, this nowcast would have been computed in April 2012, specifically 16 days after the end of the month we nowcast. The models are estimated using the vintage of TIO available in April 2012. We repeat this procedure for each month until September 2017, expanding the estimation window (instead of using a rolling window approach). This means that our estimation sample is increasing over time. As an example, in the case where we use the estimated factors as regressors we would summarize our procedure as:

$$y=\hat{F}\beta + \varepsilon \qquad (1)$$

$$\hat{y} =\widehat{F_t\beta} \qquad (2)$$

In (1) and (2), t refers to the month we want to nowcast and y and $\hat{F}$ are the TIO and estimated factors going from t = 1, . . . , t − 1. Of course the equations (1), (2) take many forms depending on the model we adopt, but the principle is similar: we first estimate the models using data until the latest month for which we have TIO values and then we use the most recent firm-level information to compute the nowcast, given the estimated model parameters.

Our quarterly estimates of GDP are entirely based on TIO, both the released version and our nowcasts. As we mentioned in the data description, TIO provides the basis for the initial estimate of GDP, hence it is optimal to use it as a regressor in a nowcasting exercise. We compute the GDP

nowcasts differently, depending on the month in which we make the estimate. In our setting, the nowcasts for a given quarter are computed three times: during the second month of the quarter, during the third month and 16 days after the end of the quarter. In the first case, we would use the nowcast of TIO for the first month of the quarter, and then estimate an automated ARIMA model (see Hyndman and Khandakar, 2008) to obtain the forecasts of the remaining months. If we compute the GDP nowcast during the third month, we would use the first TIO estimate made by Statistics Finland for the first month, then use our nowcast of TIO growth for the second month and then compute the 1-step ahead forecast for the third month. When we estimate GDP growth 16 days after the end of the quarter we use TIO growth computed by Statistics Finland for the first two months and augment them with our nowcast of TIO for the last month of the quarter. Eventually, we are going to have an estimate of TIO growth for each month of the quarter of interest and we obtain GDP growth by taking a simple average over the three months. Denote the estimate of GDP growth for quarter q going from to month t − 2 to t as $\widehat{\mathrm{GDP}}$, then our quarterly nowcast is $\widehat{\mathrm{GDP}}_{q,t} = 1/3(\hat{y}_{t-2} + \hat{y}_{t-1} + \hat{y}_t)$ Notice that this procedure is rather similar to the one of bridge regression, which links quarterly and monthly variables via simple linear models. We have tried to estimate a linear regression of GDP growth onto the quarterly average of TIO growth, i.e. estimating the linear model $\widehat{\mathrm{GDP}}_{q,t} = \beta\frac{(\hat{y}_{t-2}+\hat{y}_{t-1}+\hat{y}_t)}{3} + \varepsilon$, but our results indicate that the simple average of TIO growth is a better regressor than using the bridge formulation. The other issue that we mentioned at the beginning of this subsection concerns computational feasibility. We estimate more than 150 nowcasting models, some of which are computationally burdensome. Given that we would like to produce (and possibly release) the nowcasts around t + 16, using the information set available by then, we need to find some sort of compromise between having the largest spectrum of models and being able to estimate TIO quickly. In order to do that, we select a relatively small subset of models (around 20) which perform well on the historical sample and proceed to use these techniques to produce nowcasts for the most recent month. We then average these nowcasts using simple combination schemes such as unweighted average or using weights which depend on historical nowcasting performance (Stock and Watson, 2004, point out that these schemes outperform more complex ones). We have tried different criteria in order to trim the original nowcasting models and found that keeping the models with lowest mean error (i.e. the ones producing unbiased nowcasts of TIO) tend to produce the best TIO and GDP estimates, once combined. Once we have produced the fast estimate of the indicator of interest, we re-evaluate the whole set of models to make sure that the performance with respect to the latest months does not alter the best set of models. This implies that, in principle, the models which are going to be included in the estimate can change over time.

3.3.3 Nowcasting with traffic measurement data

As we mentioned, the traffic volumes data represent a more complicated data source compared to our set of firm-level sales. For example, they present many missing observations and the panel of measurement points needs to be constructed from the original files available on the Finnish traffic authority's webpage. Given that the data are available only from January 2010, we have decided to start the computation of pseudo-real-time nowcasts of TIO growth from January 2014, to give us four years of estimation sample. Similarly as in the firm-level data case, we adopt the predicted TIO growth rates to compute the year-on-year growth of GDP.

The traffic data are aggregated at the monthly level and we assume that our estimation of TIO is conducted around 16 days after the end of the reference month (as in the main exercise). This allows us to use the Statistics Finland's estimates of TIO for the t − 1 month, where t represents the period we want to nowcast. However, in principle the traffic data we utilize allow for nowcasts during the

month of interest, given their daily frequency. It is important to point out that, unlike the firm-level data we utilize, our set of traffic volumes contains missing values. In order to impute the missing observations, we rely on the regularized principal component technique illustrated in Josse and Husson (2016).

The R-package 'missMDA' is available that can do this procedure.

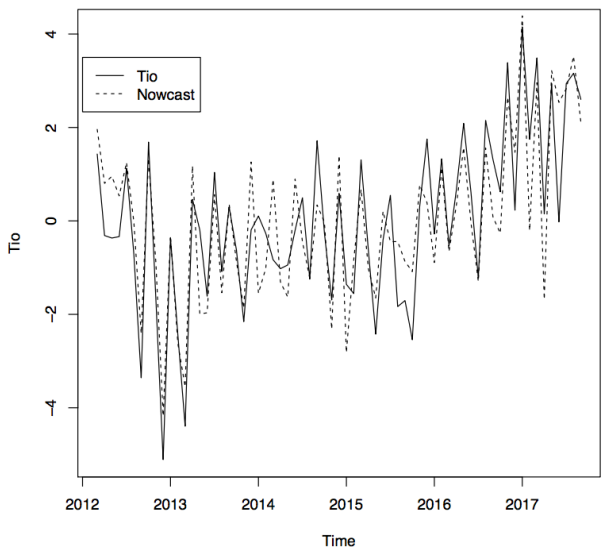
https://cran.r-project.org/web/packages/missMDA/missMDA.pdf

The actual nowcasts are computed using various statistical models and machine learning techniques. We have used a lower number of models (around 20) than in the example of firm-level sales, and the final nowcasts are obtained by making a simple unweighted average of the individual predictions, after trimming the models producing large historical mean errors.

3.3.4 Results for monthly indicator TIO

Our nowcasting exercise is centred on providing fast estimates for the year-on-year growth rate of TIO, starting from March 2012 (the first month for which we have the vintage of the data) and ending in September 2017. We now provide the results for our pseudo out-of-sample analysis. Specifically, we report the results of the models which provide the lowest root mean squared error (RMSE), the lowest mean error (ME), mean absolute error (MAE), and finally for the model with the lowest maximum absolute error (MaxE). In addition, we report the results for the simple forecast combination consisting of the unweighted average of the nowcasts provided by models with low ME. Firstly, we plot the nowcasts obtained from the forecast combination, against the first published version of TIO.

Picture 7: TIO vs Nowcast



Plots are not the most accurate tools to evaluate the performance of a nowcast model, but they do provide some intuition on the usefulness of our predictions. In this case, it seems that our firm-level data provide a good basis for providing flash estimates of TIO. The nowcasts track fairly well the original series, except for a fairly large mistake in April 2017, while they provide a substantial gain in terms of publication lag (around 30 days). Next, we provide some numerical indicators of the nowcasting performance, for the models described at the beginning of this subsection.

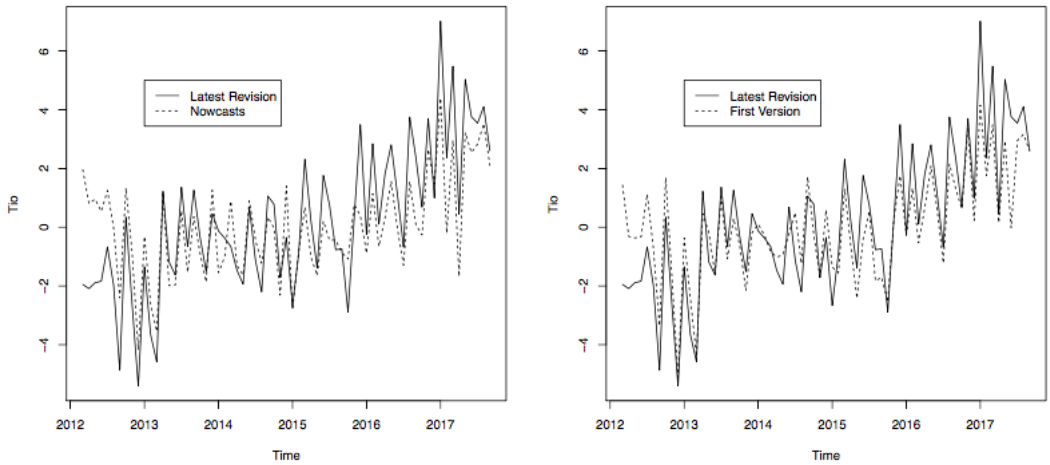
Table 13: Models that are selected based on different criteria (model with lowest mean error, lowest RMSE, lowest MAE, lowest maximum error, and combinations). The strategies are compared with the different measures of accuracy.

	Lowest ME	Lowest RMSE	Lowest MAE	Lowest MaxE	Combination
ME	0.00	-0.05	-0.05	0.00	-0.00
MAE	0.83	0.81	0.81	0.83	0.76
RMSE	1.02	0.98	0.98	1.02	0.93
MaxE	2.36	2.64	2.64	2.36	2.55

As we can see from Table 13, the nowcasting performance of the models with the lowest mean error and the lowest mean squared error is very similar. Moreover, the simple nowcast combination provides the best estimates, in terms of ME, RMSE and MAE. However, the largest error of the combination is slightly larger than the one of the lowest MaxE model. In our case, nowcast combinations seem to be the most desirable approach, also in the light of being less prone to possible structural breaks in a model's performance. Consequently, for the rest of this report, e.g. when we look at the results for quarterly GDP growth, we focus on the nowcasts obtained by combining different model predictions.

The main target of our nowcasts is the first version of TIO. This is because the later versions of the series are adjusted both for prediction errors and for additional benchmarking, meaning that we cannot be sure whether the nowcast error is due to the mistake in the prediction or because of some subsequent benchmark. However, it is still interesting to check the performance of our nowcasting framework against the final version of TIO, also because it allows us to compare our revision error against the one based on Statistics Finland's publications. We first plot the nowcasts obtained by combining the original predictions, together with the latest version of TIO. We also plot the first version of TIO against the final revision available.

Picture 8: Nowcast and first official estimate compared to the final version of TIO.



(a) TIO year-on-year growth, final version and nowcasts combination.

(b) TIO year-on-year growth, final version and first publication.

Picture 8 (a) shows a lower nowcasting performance of our approach, which is expected, given that the TIO series we use in the estimation of our model is substantially different from its later revisions. This can be seen from Figure 7 (b), where we depict the first and final version of TIO: the difference between the two series is remarkable, especially for certain periods. For example, the first official release of the year-on-year growth of TIO for July 2017 was -0.02 percentage point, which was then revised to 3.76 percentage points (interestingly, our nowcast for this month is much closer to the final value of TIO than the first release of Statistics Finland). While such extreme revisions are not common, they do show the difficulties in creating flash estimates of real economic activity. Next, we report the predictive performance measures for the nowcast combination approach, using the final value of TIO as target, even though we still use the original vintages of TIO in the estimation. We also report the same measures to evaluate the performance of the Statistics Finland's first publication.

Table 14: Comparison of accuracy of our model combination nowcasts (t+16) and Statistics Finland first publication. Both are compared to the last version of the TIO.

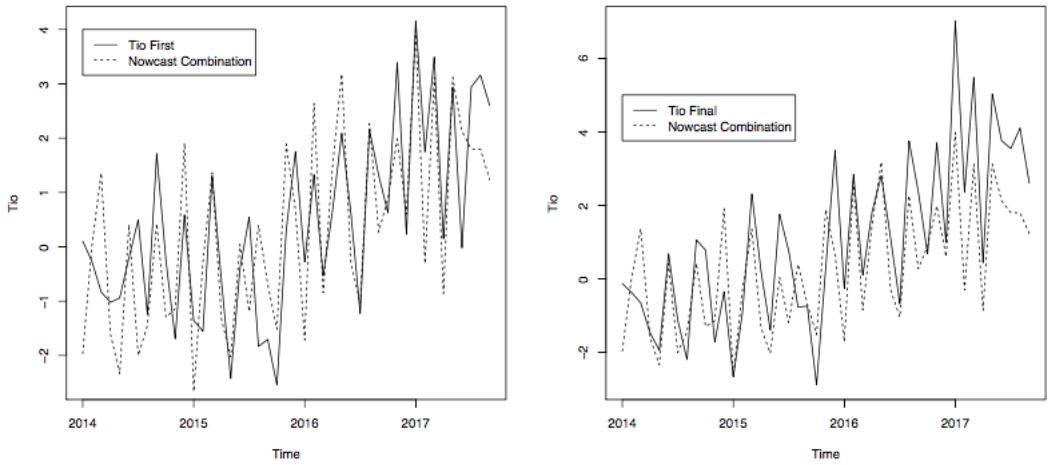
	Combination	Statistics Finland's first
ME	-0.18	-0.17
MAE	1.22	0.96
RMSE	1.48	1.22
MaxE	3.91	3.78

The performance measures reported in Table 14 confirm the fact that our nowcasting approach fairs worse when it is evaluated using the latest revision of TIO. However, it is interesting to see that the predictions of our simple nowcasting combination do not show a much larger revision error compared to the first publication of Statistics Finland (which suffers from a much longer publication lag), especially when considering the maximum absolute error.

So far, we have evaluated the performance of nowcasts based on firm-level turnovers, the core regressors of this study. However, as mentioned before we have also constructed flash estimates based on measurements of trucks' traffic volumes. First, we report the plots of the predictions

obtained by simple model combinations, where we exclude the models with historically large mean errors. We depict both the nowcasts against the first version of TIO and compared to the latest available revision.

Picture 9: Nowcast compared to the first and final versions of TIO



(a) TIO year-on-year growth, first version and nowcasts combination.

(b) TIO year-on-year growth, final version and nowcast combination.

While there are still some substantial nowcasting errors, it is impressive that an unstructured and peculiar data source such as traffic volumes is able to provide estimates that track economic activity fairly well. To gain a better grasp of how our approach is performing, we report the nowcast error measurements that we have used throughout the report, both for the first and final version of TIO.

Table 15: Comparison of accuracy of our model combination nowcasts (t+16) to the first publication and to the last

	Combination vs. First	Combination vs. Final
ME	-0.15	-0.73
MAE	1.02	1.24
RMSE	1.21	1.48
MaxE	2.50	3.02

Table 15 gives us some really interesting insights. With respect to the first version of TIO, the nowcasts combination based on traffic data provides slightly worse predictions, at least compared to the sales' data. However, the MAE and MaxE are fairly low, indicating a satisfactory nowcasting performance. When looking at the results for the latest revision we find a surprisingly small maximum absolute error, even smaller than the one of Statistics Finland's publication. Moreover, the mean absolute error and mean squared error are very similar to the ones of nowcasts based on firm-level data. The main issue with traffic data is the presence of a fairly large (in absolute terms) mean

error, indicating that our nowcasts are biased with respect to the latest version of TIO. However, we have to keep in mind the nowcasting errors obtained from the comparison with the latest revision of TIO might be caused, partially, by smoothing or benchmarking that cannot be predicted.

To summarize the results of this subsection, we have seen that combining firm-level data with statistical models and machine learning techniques that are able to deal with large dimensional datasets provide fairly accurate nowcasts, both with respect to the first and to the final version of TIO. The good predictive performance is matched with a substantial gain in timeliness, around 30 days compared to the current publication schedule. The results for the estimates based on traffic volumes evidence the potential of this data source. While the predictions are slightly worse than the ones based on firm-level data, especially compared to the first release of TIO, the errors are not extremely large. Notably, the maximum revision error obtained from this data source is even lower than the one of the first Statistics Finland's publication. The potential real-time availability of traffic data, combined with their satisfactory nowcasting performance, indicates that it is a data source that should be studied further.

3.3.5 Results for quarterly GDP nowcasts

We now turn to the results regarding the estimation of quarterly GDP year-on-year growth, in real terms. In particular, we nowcast the t + 60 release of GDP, which is the first official release made by Statistics Finland. In Section 4.1, we describe how we use the nowcasts of TIO to compute GDP growth, while this subsection is devoted to the reporting of the results. As we did for TIO, we start by plotting our nowcasts (again obtained by the simple unweighted average of the original predictions), against the official GDP growth. We do this for the nowcasts computed during the second month of the quarter, the ones produced during the third month and finally the nowcast computed 16 days after the reference quarter. The nowcasts are provided for the period going from 2012 Q2 until 2017 Q3.

Picture 10: Nowcasts of quarterly GDP compared to the t+60 publication

(a) Nowcasts during second month. (b) Nowcasts during third month. (c) Nowcasts 16 days after quarter.

Picture 10 indicates that the estimates of TIO based on our nowcasting approach provide good predictions for GDP growth, in a timely fashion. The performance of our models seems to be particularly strong when we compute the predictions during the third month of the quarter and 16 days after the end of the quarter, providing us a 45 to 75 days reduction in the publication lag. Next, we report the nowcasting performance measures for these three sets of predictions.

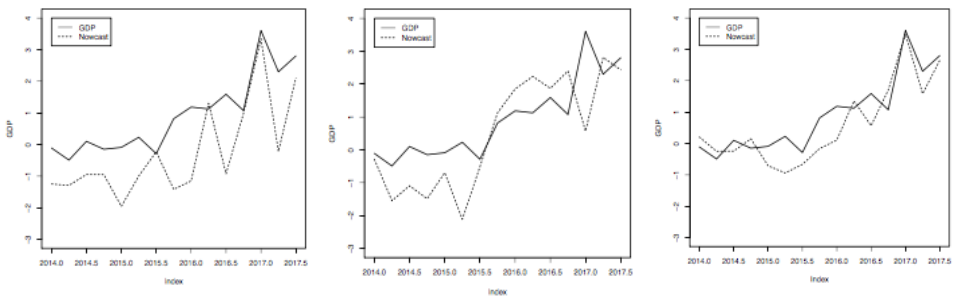
Table 16: Nowcasts of quarterly GDP compared to the t+60 publication

	Nowcast second month	Nowcast third month	Nowcasts 16 days after
ME	-0.14	-0.03	-0.12
MAE	1.07	0.66	0.49
RMSE	1.28	0.87	0.63
MaxE	2.58	2.17	1.11

Looking at Table 16, we see that our nowcasting framework is able to predict GDP accurately. Moreover, it is interesting to see that the performance of the models improves the later we computed the nowcast. This is expected, because nowcasts computed 16 days after the reference quarter rely on a much wider set of information compared to previous estimates. In particular, the latest estimates, which allow for a 45 days reduction in publication lag, present a very low MAE and a low maximum error. Overall, we can say that the nowcasts of TIO based on firm-level data are a good basis to estimate real economic activity.

Finally, we examine the performance of the nowcasts based on traffic data. We start by depicting plots similar to the ones in Picture 7, i.e. we report the predictions computed during the second and third months of the reference quarter, together with the 16 days after the end of the quarter estimates. Notice that these nowcasts go from 2014 Q1 until 2017 Q3.

Picture 11: Nowcasts of quarterly GDP compared to the t+60 publication, traffic data



(a) Nowcasts during second month. (b) Nowcasts during third month. (c) Nowcasts 16 days after quarter.

The quarterly results confirm the promising performance of traffic data for the production of early estimates of GDP. However, from the graphs it seems that the estimates computed during the second and third months of the quarter are less reliable than the ones based on firm-level data. On the other hand, the t+16 nowcasts track well GDP growth, or at least do not show a substantially different performance compared to the ones obtained through the firm-level sales. To asses in a more formal way the performance of our nowcasts, we report the error measures adopted throughout this report.

Table 17: Nowcasts of quarterly GDP compared to the t+60 publication

	Nowcast second month	Nowcast third month	Nowcasts 16 days after
ME	-1.15	-0.41	-0.32
MAE	1.18	0.97	0.55
RMSE	1.46	1.25	0.65
MaxE	2.52	3.03	1.16

The results of Table 17 confirm the intuition we gathered from Picture 11, i.e. that the nowcasts produced using traffic data have a lower predictive performance compared to the ones based on firm-level sales. This is especially true for the estimates during the second and third months of the quarter. On the other hand, the performance of the t + 16 estimates have a similar nowcasting error. Overall, it is interesting to see that traffic data are allowing us to create fairly precise estimates of GDP growth well before the official publication by Statistics Finland. Given the potentially real-time availability of traffic volumes' measurements, these results indicate the need to further explore the nowcasting ability of models based on these data.

The quarterly results reported in this subsection highlight the ability of models based on firm-level data and traffic data to provide accurate estimates of GDP growth. Even if the very early estimates, the ones computed during the quarter of reference, exhibit substantial nowcasting errors, the performance of our framework becomes significantly better when we consider the predictions at t + 16. While these flash estimates occur after the end of the quarter of reference, they allow for a 45 days reduction in the publication lag, which represents a substantial improvement.

3.4 Impact assessment of use of microdata and traffic data on estimating economic indicators in Finland

We find that using incomplete early micro-level sources and a real-time big data source such as the traffic loop data can be used to produce early estimates of economic indicators. Most notably, the work done during the SGA-1 and the SGA-2 has already led to significant methodological improvements at the Finnish statistical office. Statistics Finland regularly produces turnover index nowcasts based on the methodology presented here. Therefore, as a result of this work, it is expected that turnover indicators will become timelier and (what is remarkable) more accurate, which in turn makes it possible to produce the aggregate output indicators (TIO and GDP) faster and more accurately by the existing national accounts procedures. We also provide a new methodology by which an early estimate of GDP can be produced reliably (quantitatively with similar accuracy as the official first estimates) by employing micro-level turnovers and traffic loop data together with various machine learning methods designed to handle such large dimensional data. These results have several implications. Firstly, the statistical office can easily meet the Eurostat requirements for timelier delivery of the flash GDP. Secondly, one can potentially construct real time and possibly high frequency (e.g. daily) estimates of the economic output. Thirdly, the traffic loop data should be inspected further for instance by exploring clusters of the measurement points around designated areas (borders, manufacturing clusters, mining fields, etc.). Further, the explanatory power of traffic data should be examined in relation to the turnover indicators or other specific sectors.

4 Estimating/forecasting early economic indicators at ISTAT

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4.1 Introduction

Through an institutional collaboration agreement with the Bank of Italy, Istat had the possibility to access new data sources concerning electronic transactions of the exchange circuits and interbank settlement of the System of Payments and others deriving from the Anti-Money Laundering Aggregate Reports (SARA) that financial intermediaries file monthly to the Financial Intelligence Unit (UIF), which is a separate branch of the Bank of Italy.

A joint working group Bank of Italy-Istat was set up to produce new series of indicators from these Big Data sources, to study the characteristics of these series and to test their possible use besides traditional economic series, for the purposes of nowcasting or forecasting macroeconomic aggregates.

This report is aimed at presenting the actions taken to produce the new series and the first results of the insertion of the new series in Istat models. There are two case studies: the nowcasting of the value added of Services and the forecasting of Consumptions. The results are to be considered still preliminary and experimental.

4.2 New data sources and data access

In the System of Payments there are two main categories of payments: wholesale payments (handling large-value payments, more connected with financial markets flows and refinancing operations with national central banks) and retail payments (mostly related to individual economic activity). After 1999, the Monetary Union decided to better harmonize the infrastructure to transfer money among economic operators to foster the financial and commercial integration. Consequently, two area-wide systems were set for the wholesale payments:

- TARGET2 (Trans-European Automated Real-Time Gross Settlement Express Transfer System; T2), provided by the Eurosystem
- EURO1, privately owned

As for the retail payments, the system is not fully integrated yet. However, since 2014 the Single Euro Payments Area (SEPA) has fostered the interoperability among different national clearing and settlement retail systems.

The Bank of Italy manages the BI-Comp clearing system (in compliance with SEPA) which clears the domestic payments on a multilateral net basis. However, these payments can be settled both in BI-COMP and in TARGET2 (in case of urgency or security reasons) and this retail branch of TARGET2 is named TARGET2-retail. These data, collected on a daily basis, are disseminated by the Bank of Italy on a monthly basis with a short delay from the production; these series are not revised because they are recorded without errors by construction. Series are broken down by type of payment (transactions from direct debt, payment card, cheques and credit transfer).

The retail non-cash payments settled through BI-COMP and T2-RETAIL add up to about 60% of the total value of retail payments in Italy and 80% considering only electronic payments (Aprigliano et al, 2017).[4]

After an initial phase spent in the understanding of the specificity of these data, that is the macroeconomic phenomena associated with the series and the structure of the datasets, the Bank of Italy has carried out a census of the time series that can be useful for the goals of the project; therefore, some new and more disengaged indicators (in terms of payment instruments) have been identified and the series considered have been reconstructed backwards since January 2000.

This resulted in the production of about twenty monthly time series extracted from two inter-exchange and settlement circuits (BI-COMP and TARGET 2 retail) presented in amount and in number of transactions. The series, going from Jan. 2000 to Nov. 2017, are broken down by different payment instruments (i.e. credit transfers, direct debits, payment cards as POS and ATM, other) (Table 1). As for the coverage of these data, the series of the System of Payments concern about 40-50% of the total transactions, that is those between customers not of the same banking group.

Table 18:- Monthly time series generated by BI-COMP transactions, by Payment instruments, and by TARGET2 for the period Jan. 2000 - Nov. 2017

source	Payment instruments
BI-COMP by different payment instruments - (in number or amount)	CREDIT TRANSFER
	DIRECT DEBIT (RAV MAV, RID, SEPA DIRECT DEBIT, ETC.)
	PAYMENTS CARDS
	-- *ATM*
	-- *POS*
	CHEQUES
	OTHER
BI-COMP Multilateral balance (amount)	
TARGET2 customer payments (in number or amount)	TOTAL
	-- *CROSSBORDER*

Moreover, having understood the importance to have more disaggregated information available, from the point of view of the sectors involved in the transactions, the joint group decided to use another Big Data source deriving from the Anti-Money Laundering Aggregate Reports (SARA) filed by financial intermediaries to the Financial Intelligence Unit (UIF), a separate branch of the Bank of Italy.[5]

The database consists of anonymous data relating to all the transactions of an amount equal to or greater than EUR 15,000[6]. They are over 100 million records per year, which are collected monthly, with a delay of around T+60 days and containing information on the municipality where the transaction took place, on the kind of transaction (give/take) and summary information of the customers performing the transaction.

This activity resulted in the production of about fifty series, never calculated before, referring to the total value of the banking transactions, cash operations, domestic and foreign wire transfers. In Table B the main characteristics of the time series available from the SARA source are summarized.

[4] Aprigliano V., Ardizzi G. and Monteforte L. (2017), Using the payment system data to forecast the Italian GDP, Banca d'Italia, Temi di discussion (Working papers), Number 1098 - February

[5] Within this exercise only reports from banking intermediaries were used.

[6] In keeping with the Italian anti-money laundering regulation, some transactions below this threshold are also reported.

Table 19: Monthly time series generated by SARA System, by Payment instruments, by sector from which the payment originates and type of transaction (give / take), various periods, - Amount

SECTOR/	Payment instruments and type of transaction							
	TOTAL		CASH		BANK TRANSFERS			
					NATIONAL		ABROAD	
	Give	Take	Give	Take	Give	Take	Give	Take
TOTALE SERVICES	Since JAN. 2001							
RETAIL TRADE								
WHOLESALE TRADE								
OTHER								
MANUFACTURING INDUSTRY	Since JUN.2010							
MARKET SERVICES								
HOUSEHOLDS	Since JUN.2007							

4.3 Data handling and Data editing

As for the data access, the colleagues of the Bank of Italy or of the UIF aggregated the elementary data and produced the series in compliance with the existing rules on personal data protection.

As for the comparability over time of the series coming from the System of Payments, it should be noted that during the considered period (i.e. 2000M1 - 2017M12) these data are fully comparable except for some changes in the Regulation of the System that create some shocks. The pre-treatment and inspection phase highlighted some periods with particular perturbations for the BI-COMP series, where particularly low values are observed starting from 2013M12, due to launch of the Pan-European inter-bank exchange and settlement system for private operators (STEP2). This new private system initially has eroded the representativeness of the BI-COMP series, but this phenomenon that has greatly diminished in short term.

The SARA-UIF series revealed some perturbations in the period 2002M10-2003M12, due to problems attributable to the technology used at the time for reporting the series of Cash / Debit amounts. Thus, a preliminary phase of outliers' detections was needed to decide how to correct these breaks in the series.

4.4 Estimating/forecasting early economic indicators

4.4.1 Estimating value added of services

In Italy the early estimate of GDP is officially released at T+45 days (at T+30 days from the release of first quarter 2018), compiled using both the same sources and methods adopted for the full compilation of quarterly national accounts (released at t+60), and the same quality criteria in order to publish timely, consistent and coherent estimates of macroeconomic variables (GDP). Estimates of GDP are built from numerous sources of information, including business surveys, household and other social surveys, administrative information. Nevertheless, the availability of these sources is limited especially considering the third month of each quarter.

Currently, the main sources for estimating the value added of services are based on the following indicators: a) quarterly ISTAT turnover indices (disseminated at around T+55 days) covering market services (i.e. 38% of total services); b) administrative data coming from the ministries of Economy, Education and Health for the estimation of non-market services (21% of the value added of services); c) surveys on expenditures for household services and d) other statistics, i.e. the Bank of Italy matrix

of accounts for financial intermediation services (for a detailed analysis of the sources used, see ISTAT, 2015[7]).

In particular, no indicator related to services is sufficiently timely for rapid estimates (except for the industrial production index and the matrix of accounts for financial intermediation services) and therefore the availability of new economic series is particularly relevant. For this reason, a forecast exercise (one step ahead) of the value added of the services with advance information provided by the series of electronic transactions referring to the main payment system circuits and to the SARA anti-money laundering reports has been undertaken.

For the evaluation of the new series of payments, reference was made to autoregressive distributed lag models (ADL) and in particular that of order ADL(1,1) conducted at quarterly frequency. The formulation of the ADL(1,1) model is the following:

$$\Delta^l \Delta_4^m y_t = \phi \Delta^l \Delta_4^m y_{t-1} + \alpha + \beta_0' x_t + \beta_1' x_{t-1} + \gamma t + \varepsilon_t, \;\; \varepsilon_t \sim N(0, \sigma_\varepsilon^2), \qquad t = 1, \dots, T-1 \qquad (1)$$

Where

y_t is the target variable, in our case the value added of services in natural levels or logarithms;
Δ^l l = 0.1 the simple difference operator $\Delta^l y_t = y_t - y_{t-1}$ for the $l = 1$ or absence of differentiation for $l = 0$;
Δ_4^m $m = 0.1$ the seasonal difference operator such that $\Delta_4^m y_t$=$y_t - y_{t-4}$ for $m = 1$ and no seasonal difference for $m = 0$;
ϕ the autoregressive coefficient such that $-1 < \phi < 1$, x_t the vector of the regressors given by one or more System of Payment series to which the same differences of y_t operator is generally applied and which are considered with a lag equal to 0 or 1;
$\beta_0$and β_1 the regression coefficients associated respectively to x_t and x_{t-1};
α the constant term;
γ the coefficient associated with the linear trend t;
ε_t are the white noise residuals of the regression (assuming presence of homoscedasticity, absence of autocorrelation) whose variance is equal to σ_ε^2).

In the rolling forecasting exercise we have proceeded by means of a recursive estimate of an increasingly longer period, starting from the period 2000Q1-2011Q4 up to 2000Q1-2015Q4. Therefore, the one step ahead forecasts refer to the period 2012Q1-2015Q4, as well as the relative forecast errors which have been evaluated with respect to the most recent data release, i.e. that of the fourth quarter of 2015.

For a comparative evaluation, an exercise of pure extrapolation on data of service value added has been carried out through an autoregressive model of first order AR(1). This exercise is of crucial importance since it provides first evidences on benchmark values to which refer to in the evaluations of the new indicators in terms of mean error (ME), mean absolute errors (MAE), root mean squared errors (RMSE), etc. Moreover, it is informative on the order of differentiation, if data should be log-

[7] ISTAT (2015), I conti economici trimestrali, http://www.istat.it/it/files/2015/09/I-CONTI-ECONOMICI-TRIMESTRALI1.pdf?title=I+conti+economici+trimestrali+-+02%2Fset%2F2015+-+Nota+informativa.pdf

transformed and if the constant and trends are significant. Based on lowest MAE, main results of this exercise are as follows:

- The specification in logarithms of the data prevail for seasonally adjusted, whereas that in levels for unadjusted data.
- Concerning the difference-order, the order 1 and 4 respectively for seasonally adjusted and unadjusted data prevails.
- The constant and the trend are rarely effective.
- Seasonally adjusted data show lower values of both MAEs and RMSEs than unadjusted data.
- The lowest values of both MAEs and RMSEs are recorded for deflators, followed by the volumes and the values at current prices.

The extension of the exercise for evaluating the forecast performance of the system of payment data by ADL forms is still in progress. However, first evidences show that some series reduce the mean error statistics of the AR(1) benchmarks even if only after a careful intervention analysis for outlier removal. The reduction ranges around 12% relatively to both quarterly and annual growth rates. This gain appears in line with that obtained using traditional short-term indicators such as the industrial production index. Moreover, preliminary results of a similar exercise conducted over the service turnover index are also encouraging with gains of around 7% with respect to the AR(1) benchmarks.

4.4.2 Estimating private consumptions

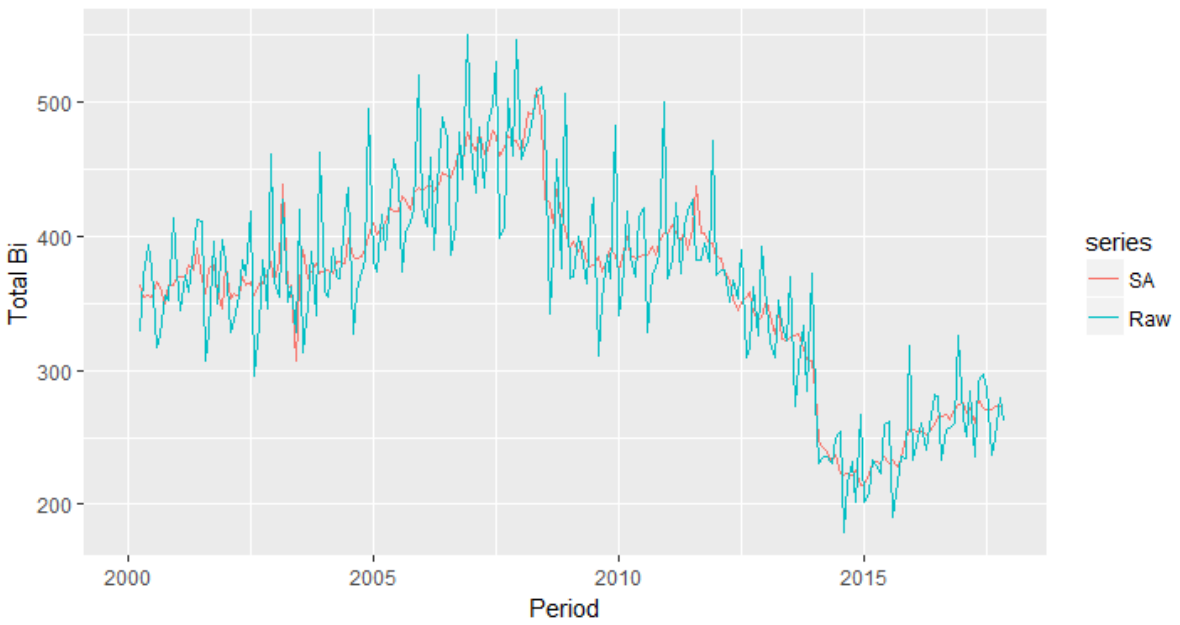
In this exercise we analyse whether the use of BI-COMP and Target2 series improves the accuracy of one-quarter ahead forecasts and nowcasts (current-quarter forecasts) of Italian private household consumption for all goods. We exploit the timeliness of data on debit card since these data are timely available and may be a valuable indicator of economic activity. We study a simple VAR model, actually used in Istat for internal forecast with the aim to find out which of the components of payments data can provide the most accurate forecast.

Data on payments need first to be seasonally adjusted. Moreover, data exhibit a change of level at the beginning of 2014 that needs to be treated using an ad hoc regressor called the ramp effect, which represents the permanent level changes that occur gradually over time. The variable used to capture a ramp effect from t_0 to t_1 is:

$$RP_t = \begin{cases} -1 & t < t_0 \\ \dfrac{t - t_1}{t_1 - t_0} & t_0 < t < t_1 \\ 0 & t > t_1 \end{cases}$$

In Picture 12 we show the results of the seasonal adjustment method performed using Demetra+. In the seasonal adjustment we tested the significance of the calendar adjustment that has been found positive using the regressor that takes into account the number of Mondays, Tuesdays,…, Sundays in the month plus the introduction of a few additive outliers.

Picture 12: Total Card Payments: raw and seasonally adjusted data

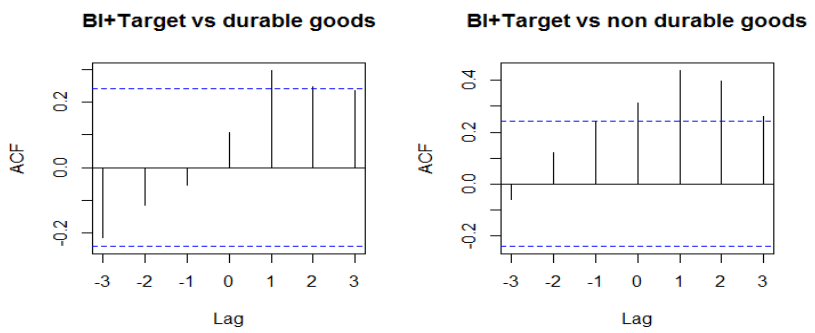


Before evaluating the value added of electronic payments data for forecasting household consumption patterns, quarterly averages have been calculated in such a way that we can calculate the cross-correlation between the components of electronic payments data and with the other variables used in the actual VAR models for durable and non-durable goods consumption. The models are separate for the two components with the following specification:

- Consumption of durable goods: Buying or building a house within the next 12 months, Car Registration, Household purchasing power
- Consumption of non-durable goods: Industrial production of consumer goods, volume of stocks currently held by retail sales companies

The cross-correlation (measured in terms on first differences) resulted particularly strong for the checks component with both the consumption of durable and non-durable goods together with the total. In such a way we start to the test if the introduction of data on electronic payments may improve the forecast accuracy. Prior to this we checked with which lag to introduce these variables in the VAR models described above. From the results in Picture 13 it is clear how data on payments have a contemporary effect for the consumption of both durable and non-durable goods.

Picture 13: Cross-autocorrelation function (SA data)



From these evidences we conduct a forecast exercise according to the following steps:

1. Each of the Bank of Italy Series has been added individually to each of the two basic prediction models
2. Series deflated with the GDP deflator
3. Verification of the significance of the parameters of the Bank of Italy series in VAR model for Consumption of durable goods and Consumption of non-durable goods
4. Forecasts one and two steps of Total Private Consumption
5. Period: 2015Q1–2017Q2
6. MAE to assess accuracy forecasts.

The parameters for the Bank of Italy variables are significant for transfers, Total BI and Target. As shown in Table 20, the total of card payments and the target variable leads to an improvement in terms of MAE both for the forecast at one step and two steps respect to the baseline model, especially the target variable performs better in both steps.

Table 20: MAE for VAR models

All Goods		
	H=1	H=2
baseline	0.00146	0.00238
BI-COMP		
checks	0.00349	0.00414
transfers	0.00160	0.00239
cards	0.00271	0.00273
cash	0.00153	0.00257
total	0.00143	0.00226
TARGET2	0.00142	0.00223

We plan to investigate further this issue considering the component of debit cards related to purchases with POS, as in preliminary studies of the Bank of Italy this component looks closely related with non-durable consumptions.

4.5 Impact assessment of use of electronic transactions of System of Payments and of the Anti-Money Laundering Reports data on estimating economic indicators

As the case studies are still exploring the use of a larger number of series and their impact on other economic indicators (i.e. turnover in service in the case of nowcasting and Investments in the case of forecasting), we consider this work is still in progress. However, at this stage, we can say that an in-depth analysis and accurate pre-treatment of the outliers of the new series is needed before these could effectively be used in nowcasting/forecasting activities. This is linked to the fluctuations showed by the series and due to the changes in the Regulations and laws ruling the sector of electronic transactions. These fluctuations are larger than those observable in commonly used short-term economic indicators. Thus, an accurate pre-treatment phase could grant to use them best and this could produce more accurate estimations of some early economic indicators.

Regarding the preliminary results of the trials, first evidences show that in the case of value added in services some of the new series reduce the mean error statistics of the AR(1) and the reduction ranges around 12% relatively to both quarterly and annual growth rates and this gain appears in line with that obtained using other traditional short-term indicators (i.e. IPI). Other similar exercises give positive results also for services turnover index. Even in the case of the forecasting model for private

consumptions some of the considered series leads to an improvement in terms of MAE both for the forecast at one step and two steps respect to the baseline model.

5 Nowcasting the ILO unemployment rate at Statistics Poland

5.1 Introduction

- We decided to nowcast the ILO unemployment rate, which is calculated on a quarterly basis and released with a **delay of four months**.
- We used a space-state model that accounts for the characteristics of the ILO unemployment rate, i.e. trend and seasonality.
- Two variables were considered: (1) *registered unemployment rate* based on data from District Labour Offices and (2) *job vacancy barometer* based on online job offers.
- We seek to answer two questions:
    - Can we precisely nowcast the ILO unemployment rate based only on the characteristics of the phenomenon?
    - Can we improve nowcast precision by accounting for the registered unemployment rate or the job vacancy barometer?

In the project we decided to use the following data sources:

- **The Labour Force Survey** – a quarterly survey conducted by Statistics Poland. The publication lag is about four months.
- **The registered unemployment rate** – calculated by Statistics Poland using data from the Ministry of Family, Labour and Social Policy. The publication lag for country level estimates is around two weeks.
- **Online job offers** – which are used to calculate the job vacancy barometer. Data were obtained using an algorithm which scraped data from 31 job offer portals. Currently, the barometer is calculated on a monthly basis but can be obtained almost in real time.

5.2 Data sources

5.2.1 Labour Force Survey

The Labour Force Survey (LFS) has been carried out in Poland as a quarterly survey called "Badanie Aktywności Ekonomicznej Ludności" (BAEL) since May 1992, according to the ILO resolution concerning the concept of "labour force surveys". It has been improved in keeping with EUROSTAT recommendations. The LFS methodology makes it possible to generalize survey results over the entire population. However, the way the LFS is currently administered (particularly its sample size) does not enable the monthly production of unemployment indicators. In each quarter, the sample consists of 4 independent elementary samples, which are partially replaced following a rotational 2-(2)-2 design (2 quarters in the survey, 2-quarter interval, again 2 quarters in the survey). Elementary samples in the LFS are selected using a two-stage stratified sampling. Statistical areas and census output areas are primary sampling units, while dwellings are selected in the second sampling stage. Currently the quarterly sample contains 55,380 dwellings. The survey covers all persons aged 15 and over who are members of sampled dwellings. Since the fourth quarter of 1999, the Polish LFS has been administered as a continuous survey. This means that quarterly samples are divided into 13 weekly samples of the same size and structure. During each of the 13 weeks of a given quarter,

interviewers visit a particular number of randomly sampled dwellings and collect data on economic activity in the previous week. Thanks to this approach, the sample reflects the situation in the labour market throughout the whole quarter.

5.2.2 Registered unemployment

One of the main sources of information on unemployment in Poland is registers of District Labour Offices (DLO). According to the current law, persons are regarded as unemployed if they are not in employment and do not perform any other kind of paid work, are available and ready to start full-time work (or in the case of disabled persons, are available and ready to start at least a part-time job); do not attend school, with the exception of schools for adults involving preparation for an extra-curriculum exam associated with the school syllabus or tertiary schools where they are attending weekend or evening classes; are registered with the District Labour Office designated for their place of residence (registered for temporary or permanent stay) and are looking for employment or other paid work. One also qualifies for the status of an unemployed person if one:

- Is aged between 18 and the retirement age;
- Is not eligible for retirement pay or pension due to inability to work;
- Does not own or hold an agricultural real estate where agricultural land exceeds 2 conversion hectares;
- Has not applied to be registered as self-employed;
- Is not temporarily under arrest or serving a prison sentence except for a sentence of imprisonment served outside a penitentiary establishment under a system of electronic monitoring;
- Does not receive monthly income exceeding 50% of the minimum wage, excluding income from interests or money kept in bank accounts;
- Does not receive a permanent social benefit or a nursing allowance or allowance supplementary to the family benefit due to single parenthood and loss of eligibility for the unemployment benefit because of the expiry of its legal duration;
- Does not receive a training allowance after their employment has been terminated.

In most cases, information in the unemployment registers is updated monthly, in others quarterly, bi-annually or annually, depending on variables involved. Data are collected at the district level (NUTS 4); as a result, any aggregation at higher levels is possible, i.e. for subregions (NUTS 3), for provinces (NUTS 2), for regions (NUTS 1) and for the whole country. Twice a year information for selected variables is available at the commune level (NUTS 5).

5.2.3 Job vacancy barometer

The job vacancy barometer is based on a web-scraped collection of online job vacancies. This index is the result of a joint project between two entities: the Bureau of Investments and Economic Cycles (pol. *Biuro Inwestycji i Cykli Ekonomicznych*) and the University of Information Technology and Management in Rzeszów. We will refer to these two entities as the data holders. The project started in 2004 and in the first period 2004-2008 the main source of data was job offers published by Gazeta Wyborcza newspaper. Thanks to the cooperation between Gazeta Wyborcza and the data holders, it was possible to include job offers from 1999 and 2003. The data collection process changed in 2008, when online job advertising services became the main source. From November 2012 several web-scraping scripts were developed in order to obtain data from 31 online services (listed below).

- http://interkadra.pl
- http://jobexplorer.pl
- http://jobexpress.pl
- http://kariera.pl
- http://karierawnieruchomosciach.pl
- http://praca.gratka.pl
- http://pracawbiurze.pl,
- http://pracawsprzedazy.pl
- http://pracownicy.it
- http://www.dlastudenta.pl
- http://www.gazetapraca.pl
- http://www.goldenline.pl/praca
- http://www.gowork.pl
- http://www.hrk.pl
- http://www.ibroker.pl
- http://www.infopraca.pl
- http://www.jober.pl
- http://www.jobs.pl
- http://www.jobs4.pl
- http://www.karierawfinansach.pl
- http://www.linguajob.pl
- http://www.metropraca.pl
- http://www.monsterpolska.pl
- http://www.mypraca.pl
- http://www.praca.pl
- http://www.pracanateraz.pl
- http://www.pracuj.pl
- http://www.profesja.pl
- http://www.przedstawiciele.pl
- http://www.strefa-hostess.pl
- http://www.szybkopraca.pl

Because the target webpages are based on various technologies (HTML5, Ajax, etc.) and have different structures (multi-page or one-page design), it is not possible to rely on existing, ready-made web-scraping tools. The web-scraping scripts were developed in Java, and are supported by the following solutions:

- **HtmlUnit**, a Java headless browser, which is used to perform HTTP requests on websites, and parse the HTML content
- **Selenium** - a webdriver, which is particularly useful when it is necessary to disguise a web-scraping bot as a human being
- **cURL** - a software project providing a library and command-line tool for transferring data from a website using various protocols, it can be used to download content from multiple URLs

Data are collected on a monthly basis at the end of each month. All job offers are downloaded and saved on a server.

Scraped data are used to calculate the job vacancy barometer, defined as:

$$\mathrm{BOP} = m_t / m_{t=2010} \times 100\%,$$

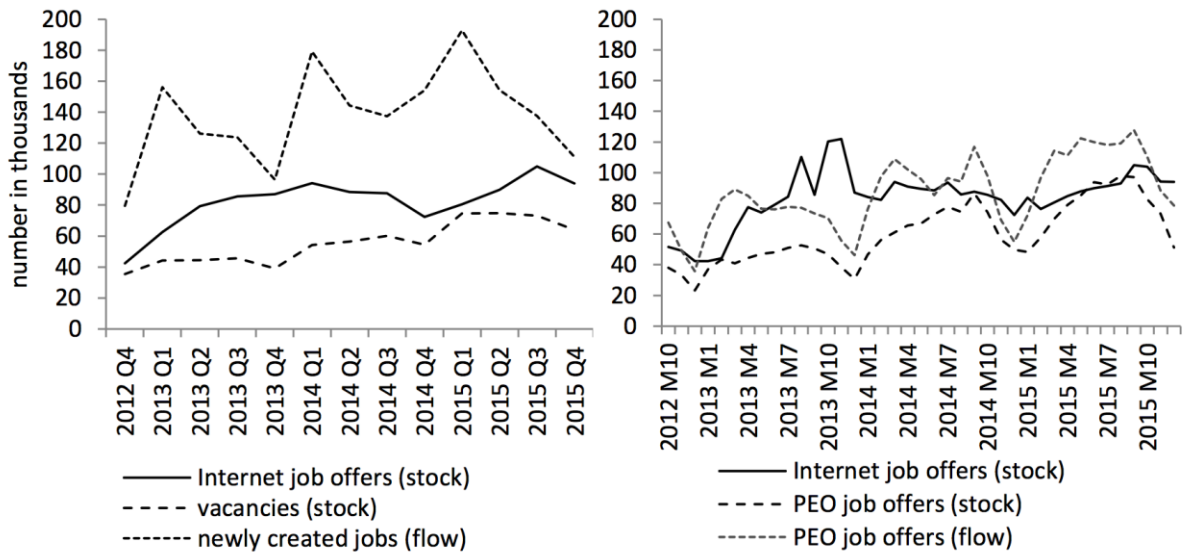
where *m* denotes the number of unique job vacancies collected.

There were some interruptions during the reference period, which made data collection impossible (June 2013, March 2015 and July 2015). They were due to sudden changes in the design of some websites or in the structure from which job offers are collected. Because of these changes the algorithm failed to appropriately locate the job offers. Therefore, for these months not all data were collected.

Between November 2012 and December 2015, on average 83,400 unique job offers at the last day of a month (with standard deviation of 18,600) were collected. The changes in Internet job offers reflected changes in the business cycle, measured by the growth rate of real GDP. In the period 2012-2014, there was a post-crisis expansion in the Polish job vacancy market. The number of Internet job offers increased from 42,400 at the end of 2012Q4 to 94,100 at the end of 2014Q1. In 2014, the real GDP growth rate stabilised. Companies reacted by decreasing demand for new workers (a fall to 72,400 job offers at the end of 2014Q4). In 2015, GDP again started to grow slightly faster, which led to a rise in the number of online offers to 93,900 at the end of 2015Q4.

Because they are collected at the end of each month, job offers do not constitute all vacancies in the economy but their specific fraction. They represent the stock of vacancies on the last day of a month. Not all job offers are included because (i) online vacancies remain posted for unknown and varying periods of time, and (ii) vacancies not posted online are not included. For this latter reason, vacancies in the public sector and job offers requiring lower levels of education tend to be underrepresented. If an advertisement is for more than one job, it is counted as one advertisement and one job offer.

Figure 14 presents three measures of vacancies: job offers scraped from the Internet (ILO, both panels), vacancies from the Demand for Labour survey (DLS, left panel) and job offers from public employment offices (PEO, right panel). The data in the picture describe either stocks - counts at the end of each month or quarter, or flows – counts for the entire month or quarter. In the period from October 2012 to December 2015, the combined stock of Internet job offers was 48% higher than the number of job offers recorded by the DLS survey (left panel) and 45% higher than the stock of job offers from public employment offices (right panel). It was 40% lower than the flow of the vacancies as measured by the DLS (left panel) and also for some periods lower than the number of job offers provided by public employment offices (right panel).



Picture 14 - Internet job offers (IJO) compared with offers from the Demand for Labour survey (DLS) and those from public employment offices (PEO). Left panel -- quarterly data on IJO and estimates based on the DLS. Right panel -- monthly data from IJO and PEOs. Stock – measured at the end of the period, Flow – measured throughout the period. Source: Pater, Szkoła and Kozak (2018).

5.3 Use of structural time series (STS) model to produce flash estimates of the ILO unemployment rate

In our study, we decided to use a structural time series (STS) model to produce flash estimates of the ILO unemployment rate. A STS model used to predict the ILO unemployment rate Y_t consists of trend L_t, seasonality S_t, regression component $X_t\beta_t$ and residuals I_t:

$$Y_t = L_t + S_t + X_t\beta_t + I_t, \qquad I_t \sim N(0, \sigma^2_{I,t}).$$

The trend component L_t is described by a local linear trend:

$$L_t = L_{t-1} + R_t,$$
$$R_t = R_{t-1} + \eta_{R,t}, \quad \eta_{R,t} \sim N(0, \sigma^2_{R,t}),$$

where R_t denotes the slope of the trend line.

The stochastic version of trigonometric seasonality is expressed by the following recursive formula:

$$S_{d,t} = S_{d,t,1} + S_{d,t,2},$$

where for $j = 1, 2$:

$$S_{t,j} = cos\left(\frac{2\pi j}{s}\right) S_{t-1,j} + sin\left(\frac{2\pi j}{s}\right) S^*_{t-1,j} + \eta_{S,t,j},$$

$$S^*_{t,j} = -sin\left(\frac{2\pi j}{s}\right) S_{t-1,j} + cos\left(\frac{2\pi j}{s}\right) S^*_{t-1,j} + \eta_{S^*,t,j},$$

$$\eta_{S,t,j}, \eta_{S^*,t,j} \sim N(0, \sigma^2_{S,t,j}).$$

In the regression component, X_t denotes auxiliary variable, β_t is the regression coefficient described by random walk:

$$\beta_t = \beta_{t-1} + \eta_{\beta,t}, \quad \eta_{\beta,t} \sim N(0, \sigma^2_{\beta,t}).$$

Residual components $I_t, \eta_{R,t}, \eta_{S,t,j}, \eta_{S^*,t,j}, \eta_{\beta,t}$ are temporally and mutually independent.

The standard procedure in the case of STS models is to express them using the state space notation (Harvey, 1989; Commandeur & Koopman, 2007):

$$\boldsymbol{Y}_t = \mathbf{Z}_t \boldsymbol{\alpha}_t + \boldsymbol{I}_t, \quad \boldsymbol{I}_t \sim \mathrm{N}(\mathbf{0}, \mathbf{G}_t),$$

$$\boldsymbol{\alpha}_t = \mathbf{T_t} \boldsymbol{\alpha}_t + \boldsymbol{\eta}_t, \quad \boldsymbol{\eta}_t \sim \mathrm{N}(\mathbf{0}, \mathbf{H}_t).$$

The first equation, referred to as a measurement equation, defines the relationship between the vector of interest $\boldsymbol{Y}_t$, and an unobserved state vector $\boldsymbol{\alpha}_t$, while the second equation, known as a transition equation, describes how the vector of parameters $\boldsymbol{\alpha}_t$ varies over time.

The estimation of parameters was conducted by the Kalman filter implemented in `KFAS` package (Helske, 2017) available in The Comprehensive `R` Archive Network.

In the study, we tested 3 models:

- model 0: without regression component,
- model 1: job vacancy barometer used as auxiliary variable X_t,
- model 2: registered unemployment rate used as auxiliary variable X_t.

We treated one-step predictions from the constructed models as flash estimates.

For assessment of flash estimates we used the following measures of accuracy:

- MAE denotes the mean absolute error:

$$\mathrm{MAE} = \frac{1}{n} \sum_t \mid \hat{y}_t - y_t \mid,$$

- MAPE denotes the mean absolute percentage error:

$$\mathrm{MAPE} = \frac{1}{n} \sum_t \mid \frac{\hat{y}_t - y_t}{y_t} \mid,$$

- MSE denotes the mean squared error:

$$\mathrm{MSE} = \frac{1}{n}\sum_t (\hat{y}_t - y_t)^2,$$

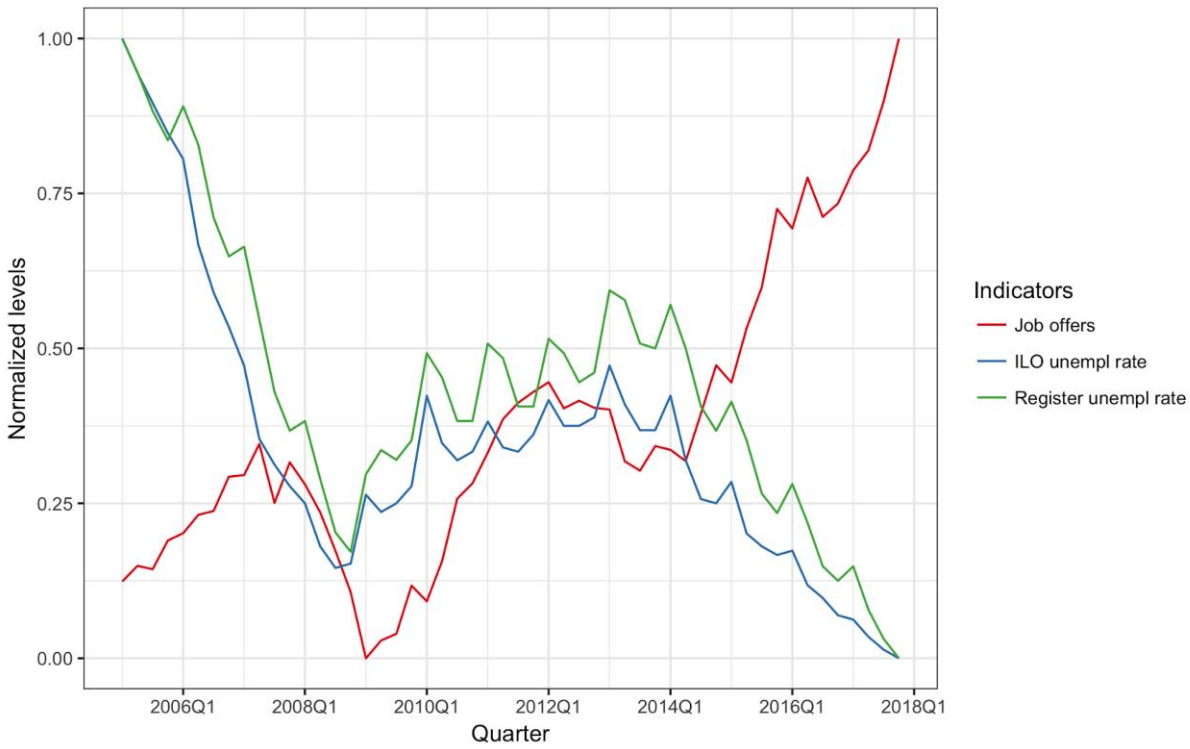
- RMSE denotes the root mean squared error:

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_t (\hat{y}_t - y_t)^2},$$

where y_t denotes observed value and $\hat{y}_t$ - the flash estimate.

5.4 Results

Picture 15 presents a comparison of the three indicators considered in the study. The red line denotes the job vacancy barometer, the blue line – the ILO unemployment rate and the green line – the registered unemployment rate. Trends in ILO and registered unemployment rates are similar in terms of both trend and seasonality. There is a rapid increase in the number of job vacancies at the beginning of 2014.



Picture 15. - Comparison of online job vacancy barometer, ILO unemployment rate and register unemployment rate

Picture 16 presents a comparison between three models considered in the study, namely model 0 without any auxiliary variables, model 1 with the job vacancy barometer and model 2 with the

registered unemployment rate. For comparison, the ILO unemployment rate is also displayed. Predictions for both models are quite similar, but all the models fail to represent the seasonal component correctly. Since 2015, a fade-out of seasonality can be observed in the ILO unemployment rate, which is only manifested by a strong decline in the trend.

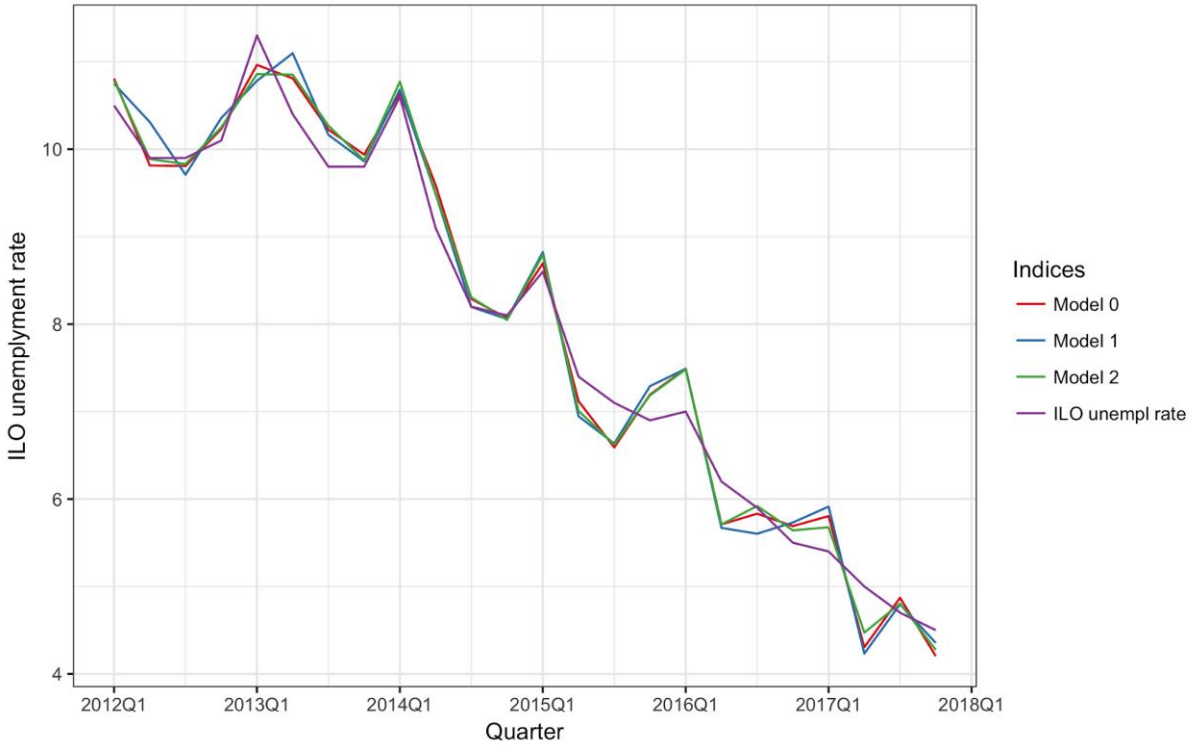


Figure 16 - Predictions from Model 0 (without any auxiliary variables), Model 1 (with job vacancy barometer) and Model 2 (with registered unemployment rate) compared to the original ILO unemployment rate between 2012 and 2018.

Based on the measures of accuracy (Table 21 below), it can be seen that the inclusion of job vacancy data as the auxiliary variable has a negative effect (compared to Model 0 and Model 2) on the accuracy of preditions.

Two possible explanations for this outcome could be proposed. First of all, it can be argued that the number of job vacancies is only indirectly related to the level of unemployment (there might be also a lot of noise in this source of data). A mere fact of an increase in the number of job offers available does not directly contribute to a decline in unemployment, since there are other factors at play. The number of job offers is an indication of the number of vacancies in the entire economy but they could just as well be filled by migrants from other countries, which would not affect the level of unemployment. At the moment, the unemployment rate in Poland is very low and many of those affected are long-term unemployed, which are structurally maladjusted to the current labour market.

Another explanation has to do with the LFS time series. The current falling trend with declining seasonality may well be the reason why random fluctuations in the vacancy barometer result in biased short-term predictions. It is possible that this indicator could perform better as an auxiliary variable if there were turning points or structural breaks in the trend.

Table 21 -- Measures of accuracy

Measure	Model 0	Model 1	Model 2
MAE	0.2736	0.3281	0.2624
MAPE	0.0389	0.0451	0.0357
MSE	0.1076	0.1482	0.0977
RMSE	0.3280	0.3850	0.3126

Note: Mean absolute error (MAE), mean absolute percentage error (MAPE), mean squared error (MSE) and root mean squared error (RMSE).

5.5 Summary of results

- The job vacancy index actually worsens the ILO unemployment rate.
- The register unemployment rate slightly improves the prediction of the ILO unemployment rate. However, one should take into account that we analysed data at the national level. From previous studies we know that prediction is better at lower levels of aggregation.

The model without any variables seems to be sufficient but this is mainly because of a clear **falling trend** in the unemployment rate.

6 The use of high frequency indicators for predicting macroeconomic variables: the experience of Statistics Portugal

6.1 Introduction

Economic forecasts are often made toward predicting quarterly or annual GDP growth rates. The goal is sometimes to identify and qualify the estimation of a measure of a given short-term economic indicator for a given reference period in a specific point in time. Due to the recent availability of high frequency / high volume data, many organizations are in search of integrating Big Data in the production of regular statistics, namely in the prediction of macroeconomic indicators. We are aware that the construction of higher frequency measures does not automatically imply an improvement in the timeliness of the low frequency corresponding ones. However, it is known that the availability of more frequent observations (in time and in space) can be considered as an improvement in terms of the econometric models used to anticipate the economic situation. In this chapter we focus on the experience of Statistics Portugal, where we applied panel data models to predict the regional GDP based on three high frequency variables.

6.2 Data sources

In Statistics Portugal, we aim at assessing early estimates for macroeconomic variables of interest over the most recent reference period based on high frequency data in the regular production of official statistics. More specifically we want to anticipate estimates of regional GDP, economic climate, consumer price index, retail sales, etc., using available high frequency data.

The regional GDP at NUTS 3 level (by sector: industry and internal commerce) is the economic indicator that is used as the predicted variable. Values for regional GDP by economic sector are not updated regularly (last data available is from 2015), so it is important to study alternative ways to anticipate estimates for these indicators. One limitation is that the time granularity of these variables is low, since GDP is a quarterly indicator. However, we may argue, again, that more frequent observations used as independent variables can always be considered as an improvement in terms of the econometric models used to produce early estimates of regional GDP.

The volume of exports (INE, 2018), overnight stays, and the withdrawals on ATMs. collected at NUTS 3 level are used as regressors (predictors):

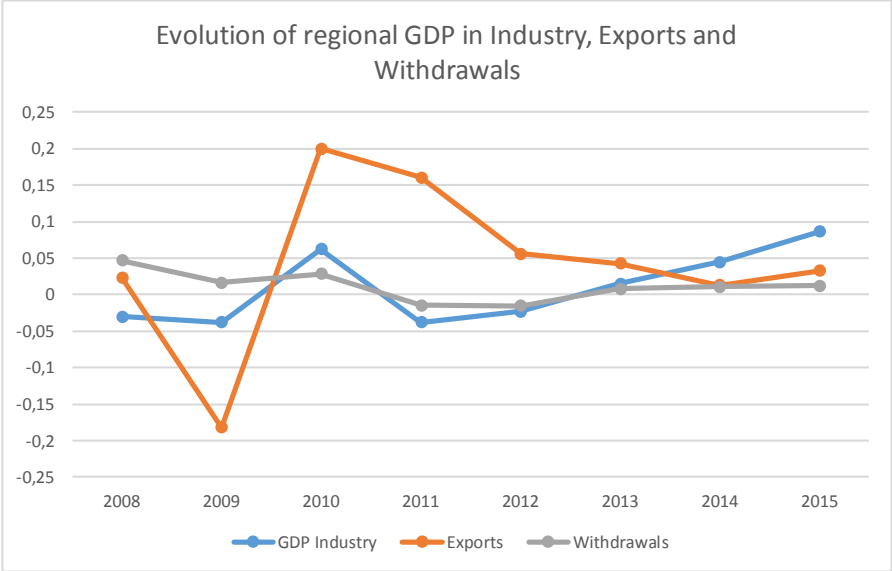
- Exports are collected by Statistics Portugal based on Intrastat, the system for collecting information and producing statistics on the trade in goods between countries of the European Union. This information has been collected monthly and by municipality. We consider that this is an important regressor for predicting Industrial GDP.
- Overnight stays are also collected by Statistics Portugal through a survey that provides information in a monthly base by municipality. We believe that overnight stays is an important regressor for predicting the GDP component of Internal Commerce.
- Withdrawals on ATM is provided to Statistics Portugal by SIBS, the Interbank Services Provider, through the Credit institutions and financial corporations survey. SIBS manages the MULTIBANCO ATM Network, one of the largest integrated interbank networks in Europe, covering more than 12,700 terminals, monthly processing more than 75 million transactions, and automating many banking functions allowing cardholders from several international brands. This information is available by month and by municipality. We believe that this indicator may be used to predict both components of GDP: industry and internal commerce.

These three variables are collected by month and are available from 2007 to 2017. These are high frequency indicators, in the sense that they can be estimated in a continuous manner. One can imagine that information on overnight stays can be collected in periods of 15 minutes, or even smaller periods. The same applies to withdrawals. However, in a first step, all the information has been aggregated by NUTS 3 in order to match the temporal aggregation of the regional GDP. Since GDP is available from 2010 to 2015, this will be the time scope of the analysis. Yearly estimations will be provided.

6.3 Econometric Modelling

Regression, Panel Data Models, and Spatial Panel Data Models are used as methodological approaches. According to Hsiao (1986), panel data require a large number of observations, since variables are being evaluated over time. Therefore, a reasonable trade-off is needed between the spatial and time granularities. The model is defined at yearly terms (2007-2015), for 23 different units of NUTS 3. For a matter of illustration, we show the overall evolution of variation rates of Industrial GDP, Exports and Withdrawals between 2008 and 2015.

Evolution of variation rates of Industrial GDP, Exports and Withdrawals



It seems that Industrial GDP has a positive, somewhat strong correlation with withdrawals (the same being not exactly true when considering the Exports)

Although the correlation between the variables is fairly good in some NUTS 3, this is not always the case in every Portuguese regions NUTS. Anyway, it seems reasonable to try a panel data model with these three variables.

We start by testing two different (alternative models): one with fixed and one with random effects model. To verify which of the previous models best fits our data, Greene (1997) suggests the Hausman test, that we apply later on.

Fixed Effects Model

The Fixed Effects Model aims to control the effects of omitted variables that vary between individuals and remain constant over time. It assumes that the intercept varies from individual to individual, but is constant over time. According to Hill, Griffiths and Judge (1999), the model is formulated as follows:

$$y_{it} = \alpha_i + \beta_1 x_{1it} + \cdots + \beta_k x_{kit} + \varepsilon_{it}$$

where,

y_{it} represents the ith observation of the dependent variable at instant t.

α_i is the estimated intercept for the ith observation

β_j is the regression coefficient: $j \in \{1, \dots, k\}$

ε_{it} represents the residual of the ith observation at instant t

As the response parameters do not vary between individuals nor over time, differences in behaviour between them will be captured by the intercept, α_i . Thus, this intercept is interpreted as the effect of the omitted variables in the model. In the Fixed effects model the intercept is a fixed and unknown parameter that captures the differences between the individuals in the sample, so the inferences made about the model are only about the individuals from whom the data is available.

To test whether the assumptions of the model are valid it is important to test to see if the intercepts are different between individuals. The null and alternative hypothesis suggested by Hill, Griffiths and Judge (1993) are:

$$H_0: a_{01} = a_{02} = \ldots = a_{0k}$$

$$vs$$

$$H_1: intercepts\ a_{0i}\ are\ different$$

These hypotheses can be tested using a F test.

The fixed-effects model is the best option for modelling panel data, where the intercept, αi, is correlated with the explanatory variables in any time period. In addition, since the model intercept is treated as a fixed parameter, it is also preferable to use fixed effects when observations are obtained from the entire population (Wooldridge, 2002)

Random effects model

The random effects model has the same assumptions as the fixed effects model, that is, the intercept varies from one individual to another, but not over time, and the response parameters are constant for all individuals and in all periods of time. The difference is the intercept.

As suggested by Hill, Griffiths and Judge (1999), unlike the fixed effects model, the random effects, does not have the fixed parameter as the intercept. The n (be the number of individuals present) intercepts are modeled as follows:

$$\beta_{0i} = \overline{\beta_0} + \alpha_i, i = 1, .., n$$

We note that this intercept is calculated according to the intercept of the fixed effects, which captures the behavioral differences of the individuals, and by $\overline{\beta_0}$, which corresponds to the population intercept. Thus the model, is represented in the following way:

$$y_{it} = \overline{\beta_0} + \beta_1 x_{1it} + \cdots + \beta_k x_{kit} + v_{it}$$

Where $v_{it} = \alpha_i + \varepsilon_{it}$ represent the errors.

To verify which of the previous models best fits our data, Greene (1997) suggests the Hausman test and has as null and alternative hypotheses:

H_0: α_i is not correlated with explanatory variables
vs
H_1: α_i is correlated with the explanatory variables

If H_0 is not rejected, there is no statistical evidence that α_i is correlated with explanatory variables. Therefore, the random effects model should be used. On the other hand, if you reject H_0, you should use the fixed effects model. In panel data analysis, the Hausman test can help to choose between a fixed effects model and a random effects model. Essentially, the tests look to see if there is a correlation between the unique errors and the regressors in the model.

SAR / SEM Models

SAR — The Spatial Autoregressive Model, says that levels of the dependent variable y depend on the levels of y in neighbouring regions (Viton, 2010). It is thus a formulation of the idea of a spatial spill over. The model is formulated as follows:

$$y = \lambda W y + X\beta + u$$

In SEM — The Spatial Error Model, the spatial influence comes only through the error terms: we have

$$y = X\beta + u$$

$$u = \rho W u + v$$

With v assumed to be normal with $E(v) = 0$.

To decide whether to use SAR or SEM models, one can use the Moran index, or the Lagrange multiplier diagnostics for spatial dependence. If there is dependence at the level of the dependent variable, we will consider a SAR, otherwise it will be a SEM model. The Lagrange multiplier test has as the following hypotheses:

H0: spatial independence vs.

H1: spatial dependence.

The Lagrange Multiplier can be used both to detect possible spatial dependence of GDP, and also to detect dependence on the model errors.

6.4. Results

We have considered two different models:

Model 1. The first model has the GDP (Internal Commerce) as dependent variable. Taking into account the dependent variables, we have run different alternatives, and found out that the best alternative uses withdrawals as the dependent variable. Overnight stays was not considered as a good predictor of GDP (Internal Commerce), as the behaviour of the series is unstable during the period. *Internal Commerce GDP ~ withdrawals*

Model 2. The second model considers the GDP (Industrial) as dependent variable. In this case, we used both exports and withdrawals as regressors:

Industrial GDP ~ withdrawals+exports

Model 1

In Model 1, and according to the Hausman Test, used to determine which model is the most appropriate, we concluded that the fixed effects model is preferable than the random effect model. In order to detect a possible spatial dependence of GDP, we used the Lagrange multiplier diagnostics for spatial dependence. The Lagrange Multiplier can be used both to detect possible spatial dependence of GDP, and also to detect dependence on the model According to the results, we conclude that there is no spatial dependence of the GDP, but errors are correlated errors (p-value=0.049). However, the SEM cannot be applied since the test for the correlation of errors does not allow rejecting H0. Therefore, we choose to apply a non-spatial panel data model. It contains the fixed part, but not the spatial component.

Model 2

For Model 2, and according to the Hausman Test, used to determine which model is the most appropriate, we concluded that the random effects model is preferable than the fixed effects model. The Lagrange Multiplier is inconclusive and, again, the SEM and SAE do not apply in this case. Therefore, we also apply a non-spatial panel data model.

For both models we have computed the mean absolute error (MAE) and the the root mean squared error (RMSE) given by the following equations:

$$\mathrm{MAE} = \frac{1}{n}\sum_{t} \mid \hat{y}_t - y_t \mid,$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_t (\hat{y}_t - y_t)^2},$$

where y_t denotes observed value and $\hat{y}_t$ - the flash estimate

Table 22 show the results where we can observe that the accuracy is still low: MAE is greater in model 1, but RMSE is greater in Model 2. For a matter of illustration, we show the values of

Table 22 -- Measures of accuracy

	MAE	RMSE
Model 1	132	3179
Model 2	1335	2368

6.5. Conclusion

In this chapter we provided the experience of Statistics Portugal in the estimation of the regional GDP at NUTS 3 level by sector: industry and internal commerce. Values for regional GDP by economic sector are not updated regularly (last data available is from 2015), so it is important to study alternative ways to anticipate estimates for these indicators. In our experience

, we ran different panel data models and concluded that the spatial effects are not important. Predictions have some errors and, therefore, we need to develop new models and use other regressors to improve the quality of our estimates.

7 Conclusions

In the Slovenian and Finnish setting, the traffic loop data are open to the general public, while the enterprise data are collected for the purpose of official statistics production and protected by the strict confidentiality standards of the statistical office. However, similar data collections exist in the other statistical offices of most countries, making our proposed approach and data source an interesting possibility for data users who need timely information on the state of the economy. Statistical offices have the possibility to increase their own relevance as information producers by using this kind of novel techniques. The relatively small investments that are required are related to modelling skills (in maintaining and updating the models) and adding a few features in the existing IT systems for storing information on the models, results and source data.

There are some things to consider in the future. Namely, the fact that some series are subject to revisions makes the choice of the correct version important. The training data might change according to what we want to estimate (the early values or the later values) and which period is being estimated.

Next, to increase the accuracy more, we might consider calculating estimations only on some components of the target series, which is to say on those that are still unknown when we are producing our predictions. Using them in conjunction with known components to calculate the target variable in a 'natural' way should yield better results, with errors on a smaller scale.

8 Literature

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